

# Response to Load

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In our last chapter, we explored the concept of training load as it relates to HIIT. However, many coaches and practitioners often get confused with a related, but independent, concept: the response to training load. As these are exclusively different concepts, we need a unique set of tools and understanding to have impact. Importantly, our athlete's load *response*, commonly termed *training status*, is not uniform across all athletes and clearly depends on the training and individual characteristics. Recall from figure 8.2 that the training load (internal and external) can be considered the INPUT, whereas the response to the load can be considered the system's OUTPUT. Providing meaningful surveillance over an athlete's unique response to training, and especially HIIT, has been considered the holy grail of coaching and applied sport science. Just as with internal and external markers of training load (chapter 8), development of microsensor technology has equipped scientists and practitioners with a vast array of tools for describing this response to training load. We have many at our disposal, and a secondary purpose of this chapter is to describe those we find useful for the practitioner.

As we just learned in chapter 8, training load, the theoretical input marker of training stress (chapter 7), can be considered from both an external (mechanical work) and internal (metabolic, cardiovascular, neu-

romuscular) standpoint (figure 8.2). While we can use these load concepts to program training intelligently, they tell us little with respect to how exactly athletes have reacted or responded to HIIT, the output, and where an athlete actually lies from a readiness-to-train perspective. Without somehow gaining insight into the output response, we're really just guessing as to what's going on in the athlete. It's precisely via the monitoring response to the training load that we can assess fitness and readiness to perform. We need tools therefore to gain insight into the output response.

In simple terms, insight into fitness and readiness to train will be gained by examining:

- Markers of fitness, fatigue, and health, which taken together may inform on performance capacity, as well as
- Efficiency or cost/output models, such as the ratios between the internal and external responses to training, in which the lower the ratios, the greater the performance capacity.

As highlighted in chapter 7, for any type of stress, the body's survival response is the process of adaptation or putting defense systems in place so that it may better remain in homeostasis should the stressor be encountered again. As we've emphasized from

the beginning, we all respond uniquely to any form of stress. This unique response provides the opportunity to modify the training content through programming to keep athletes on track and progressing toward their performance goals. Therefore, this unique response to training forms the final step in the process to help us solve our programming puzzle. Understanding an athlete's unique response to training can help us to:

- understand the athlete's individual response to training, i.e., training status;
- estimate fitness, fatigue, and performance potential;
- determine whether athletes are adapting to their training programs, i.e., gaining fitness or assessing fatigue and associated need for recovery; and
- minimize risk of nonfunctional overreaching, injury, and illness (chapter 7).

The aims of this chapter then are to:

1. highlight current training response markers that appear useful to monitor for both individual and team sports,
2. describe the current and ever-evolving challenges that practitioners face when monitoring an athlete's response to training and health status,
3. offer a framework, including simple statistics, that may be used in an applied setting to assess meaningful changes in training status, and
4. provide guidance on how to improve data visualization and increase coaching staff buy-in for sport scientists embedded in larger high-performance programs.

Combined through the diverse team of authors assembled, our aim is to provide readers with the opportunity to improve their ability to make informed decisions that assist them when solving their training program puzzles, with particular reference to HIIT, and positively impact an athlete's future performance. The overarching concepts that gain us insight into fitness, fatigue, and readiness to train and perform are illustrated in figure 9.1. As the figure shows, potential performance capacity, or training status, can be inferred using direct or indirect measures. Direct markers include races and competitions, fitness and strength tests, as well as maximal efforts in training (including HIIT, such

as key sets), whereas indirect markers of performance capacity include fitness and fatigue responses, and internal-to-external load ratios. We will begin by outlining the main indirect training response category tools that we will discuss in the chapter.

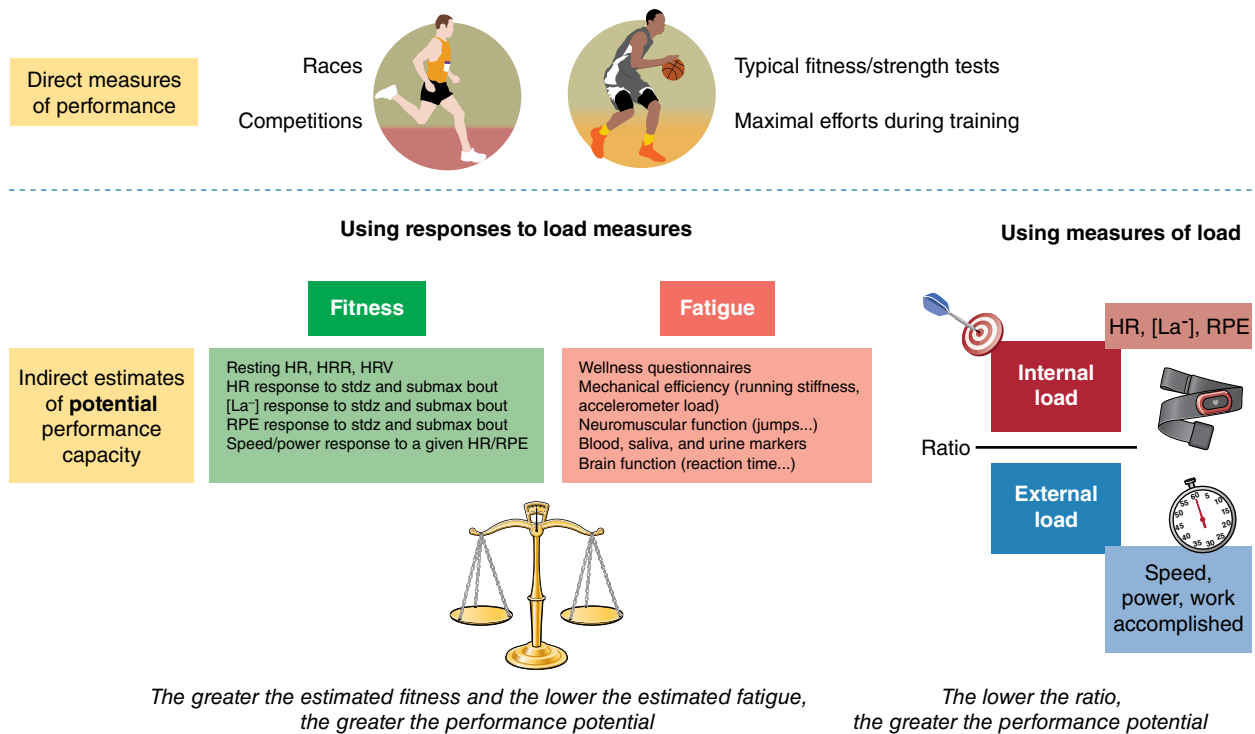
## Training Response Markers

As discussed throughout chapter 8, athlete tracking is considered one of the most important initiatives for both individual and team sports. For the weapons we implemented in training (chapters 4, 5, and 6), how lethal were they in terms of the damage they inflicted in our athletes? We need to understand this damage response to the applied training load weapons, or the carryover effect of the load. These incidences occur after the load has been applied, providing insight into the body's response to training in the period following (hours to days after). These responses provide insight into an athlete's performance level without directly measuring it.

Training responses in athletes are generally related to the training stimuli (e.g., relative or internal training load; chapter 8) (16). While both too much or not enough training load can contribute to fatigue (nonfunctional overreaching or overtraining; chapter 7), the appropriate training dose at the individual level may allow for optimal improvements in fitness and performance (16, 20, 55, 76, 78, 115, 116). It is therefore vital that an athlete's fatigue, fitness, and/or performance responses during the various training phases of an athlete's periodized calendar are under surveillance, so training load and its contents can be adjusted and individualized, both during and between training cycles (16, 97, 142, 159).

When it comes to monitoring athlete fatigue and/or performance responses to training load, various tools and methods can be used, such as the monitoring of saliva and specific blood variables (82, 121), the use of psychometric questionnaires (16, 47, 83, 153), and of course, generally knowing the athlete on a personal level and regularly communicating with him (remember to never treat a stranger; chapter 7). Changes in blood lactate can be used to track changes in fitness (chapter 8) (11), but its usefulness to inform on fatigue is, in our opinion, unlikely (109, 121). Moreover, in addition to being painful and inconvenient for frequent monitoring, the actual value of blood sampling is also questionable (81, 123).

We believe there are primarily three distinct families of key importance, each related to specific aspects of training status, that may all be useful in



**Figure 9.1** Best practice determinants of potential performance capacity. Performance capacity, or training status, can be inferred using direct or indirect measures. Direct markers include races and competitions, fitness and strength tests, and maximal efforts in training. Indirect markers of performance capacity include fitness and fatigue responses and internal-to-external load ratios.

assessing performance potential or readiness to train. Alongside experts in these respective areas, we herein describe each of these three categories with the tools we believe best represent them. Globally, and very simply for the practitioner, these include:

1. Internal metabolic, cardiopulmonary, and autonomic status, with heart rate the tool chosen to infer on the status of that system;
2. Neuromuscular and external force production and locomotor efficiency using a global positioning system (GPS) and accelerometers;
3. Overall health, fatigue, and mood (psychology); wellness questionnaires are a simple means of gaining insight into this important (and global) aspect.

## Tool 1: Heart Rate Measures

**With Jamie Stanley, Daniel Plews, and Hani Al Haddad**

Today, there is high interest in the ability of noninvasive surveillance tools to monitor divisions of

biology and physiology. Arguably the key player in the physiological domain is the autonomic nervous system (ANS), the body's central controller of, and responder to, all physiological systems, whose main role is homeostasis and balance maintenance. Conveniently for the practitioner, the ANS has a direct influence on the heart rate (HR), which can be observed through the level and variability of HR at rest (142, 159), following exercise (43, 45-47), during exercise (43, 46, 70, 147), and during recovery after exercise (59). The measurement of heart rate is noninvasive, inexpensive, time efficient, and can be applied routinely and simultaneously in large numbers of athletes. While the collection of beat-to-beat heart rate was initially only possible with expensive laboratory-based electrocardiograph recorders, the recent availability of valid and portable recorders, such as heart rate monitors (68, 167, 168), specifically designed systems (54, 135), and smartphone applications (66, 145) has substantially boosted the use of heart rate variability (HRV) monitoring in the field.

In this section, we specifically describe how heart rate measurements can be used successfully to

monitor acute fatigue and recovery responses to isolated periods of endurance-based training, which allow us to adjust training load daily, and both positive and negative adaptations to blocks of endurance-based training. The application of each measure, together with statistical guidelines to assist interpretation, is outlined along with a simple decision process to assist practitioners in selecting the heart rate variables most suited to their needs.

## One Simple Variable, Multiple Complex Indices

The measurement of a person's heart rate is simple and straightforward. It can be measured during sleep, upon waking, during exercise, or in the period following exercise. Analysis of heart rate can be as simple as reporting a 20 s value or as complex as delving into the beat-by-beat variance, more commonly known as the HRV. All methods and conditions have their advantages and disadvantages, with different determining factors (36), adaptation time courses (46), and sensitivities to fitness (43, 46, 47). For example, heart rate-derived indices such as exercise heart rate (H<sub>Re</sub>) and post-exercise heart rate recovery (H<sub>RR</sub>) can occur relatively rapidly (<1 wk after the start of an endurance training program), yet changes in resting HRV can take much longer to manifest (i.e., 4 wk of training (46)). While all measures of heart rate relate to ANS activity, the findings suggest that different indices provide insight into diverse physiological aspects of adaptation to training load and HIIT (30, 36, 46). Their use in combination therefore might improve surveillance of individual athletes. Although the appropriate combination, timing, and data collection methods need determining, it has become clear that the regular monitoring of a few simple markers can substantially improve the training process (46, 142). In the following section we present the physiological determinants and practical usefulness of each heart rate measurement.

## Resting Measures

The physiological determinants, reliability, and sensitivity of the different heart rate measures are shown in table 9.1. The basic principle behind monitoring such indices repeatedly over time relates to our attempts to understand the cardiac ANS status in response to training. Since ANS activity is highly sensitive to environmental conditions (e.g., noise, light, temperature (1)), it is important to standardize recording conditions to isolate the training-induced

effects on the ANS. While evening (nocturnal) recordings theoretically represent the best (more standardized) recording condition for HR(V) monitoring (21, 138), the effects that differences in sleep patterns (132) and quality (51) have on HRV, independent of training-related changes in ANS status, can unfortunately result in misinterpretation (figure 9.2). To overcome such limitations, HRV data measured during select slow wave sleep (SWS) episodes, which offers the best signal stability, might be preferred (21, 40). However, a typical day's activity level tends to affect the night HRV during the early hours of sleep, when the higher proportion of SWS occurs (127). As will be detailed, consideration of the day's training load that precedes the measure is crucial to correctly interpret the changes in ANS status interpreted from HRV recordings.

Currently, the best practice for athletes tends to be short-term (1-5 min) measurements of HR(V) after waking in the morning (figure 9.3) (64, 128, 142, 159). While both supine and standing recordings are often used in the literature (156), the supine condition is better tolerated by athletes and is simple to do in the morning after waking, especially using smartphone applications that use the camera lens (photoplethysmography) to measure heart rate through the fingertips, a method that has been validated against chest strap and echocardiogram (145). These morning resting measures overcome the limitations of overnight recordings while still offering a level of standardization (e.g., same bed and same time most days, quiet environment, without the immediate effect of daily activities). Measurement using such systems is simple, convenient, and therefore highly practical for athlete and team practitioner.

Surprisingly, while morning resting heart rate has been used for decades by coaches, scientists have essentially focused their research on morning resting HRV, considering it to be a more powerful tool than heart rate. However, the possibly greater sensitivity of HRV is unlikely as large as previously thought. Indeed, we recently showed only a slightly stronger trend toward nonfunctional overreaching detection from vagal-related HRV indices over morning resting HR ( $r=0.88$  vs.  $0.81$ ) (139). Similarly, the relationship between changes in resting HRV and 10 km running performance ( $r=0.76$ ) were only slightly greater compared to changes in HR ( $r=0.73$ ) (141).

There is no shortage of methods for assessing HRV (161). The most common methods include time domain and spectral analyses, while nonlinear methods such as entropy and symbolic analyses have also

**Table 9.1** Determinants, Reliability, and Sensitivity of Different Heart Rate Measures

|                                         | Determinants                                                    | Monitoring variable(s)                                                                                                                  | Typical error, expressed as coefficient of variation                          | Signal-to-noise ratio* | Smallest worthwhile change* |
|-----------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------|-----------------------------|
| Resting HR                              | Cardiac morphology, plasma volume, ANS, and baroreflex          | Wellness, fitness, readiness to perform                                                                                                 | ~10%                                                                          | 0.7                    | ~-2%                        |
| Resting vagal-related HRV indices       | Genetics, plasma volume, ANS, and baroreflex                    | Wellness, fitness, readiness to perform                                                                                                 | Index-dependent, e.g., ~12 (Ln rMSSD) to ~80 (LF/HF) %                        | 0.8                    | ~+3%                        |
| Exercise HR                             | Fitness, plasma volume                                          | Aerobic fitness                                                                                                                         | ~3%                                                                           | 1.6                    | ~-1%                        |
| Exercise HRV                            | Intensity-dependent: ANS < VT1, respiration > VT2               | In theory, aerobic fitness                                                                                                              | Index-dependent, e.g., ~60 (Ln rMSSD) to ~150 (LF/HF) %                       | N/A                    | N/A                         |
| Post-exercise HRR                       | Theoretically ANS and genetics but essentially metaboreflex     | In theory, wellness, fitness, and readiness to perform. In practice, more fitness because of its link with relative exercise intensity. | Index-dependent, e.g., ~25 (HRR <sub>60s</sub> ) to ~35 (HRR <sub>τ</sub> ) % | 1.3                    | ~+7%                        |
| Post-exercise vagal-related HRV indices | ANS and baroreflex, but the metaboreflex has the greater effect | In theory, wellness, fitness, and readiness to perform. In practice, more fitness because of its link with relative exercise intensity. | Index-dependent, e.g., ~16 (Ln rMSSD) to ~65 (LF/HF) %                        | 1.1                    | ~+4%                        |

HR: heart rate; HRR<sub>60s</sub>: HR recovery within 60 s following exercise; HRR<sub>τ</sub>: time constant of HR recovery derived from monoexponential modeling; HRV: HR variability; N/A: not available. Typical error: see text for related references. Signal-to-noise ratio refers to the magnitude of the change observed in (31) divided by the typical error reported in (4). See text for the methods used to calculate the smallest worthwhile change. \*: for positive training adaptations.

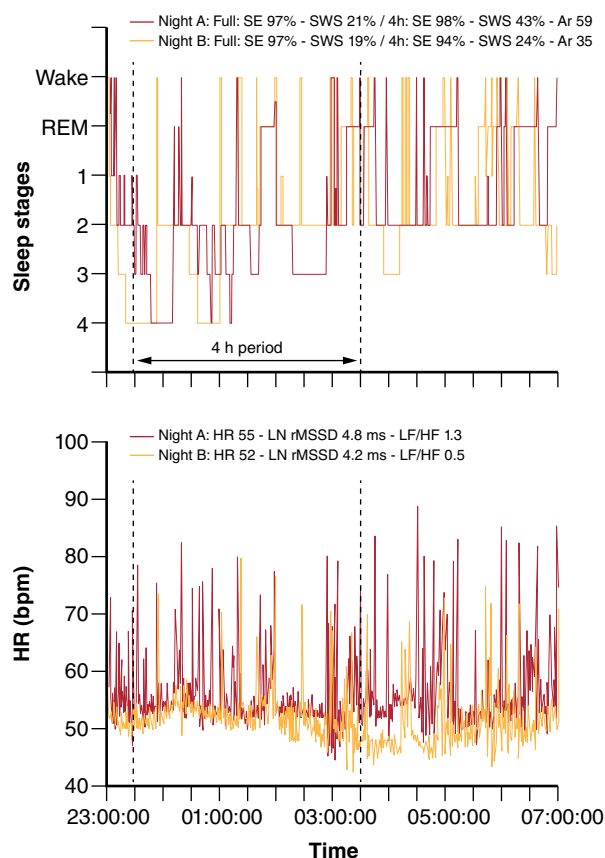
gained some interest (161). Each index captures a different feature of the ANS, with some indices more likely to reflect cardiac sympathetic activity and others cardiac parasympathetic activity (161). In practice, however, when it comes to selecting the more appropriate HRV indices to monitor athletes in the field, time domain indices (e.g., rMSSD (square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals) or SD1 (standard deviation of instantaneous beat-to-beat R-R interval variability measured from Poincaré plots), both reflecting parasympathetic modulation, appear to be the best. These measures can be captured reliably over a very limited time (e.g., 10 s to 1 min (74, 130, 131)) and are compatible with the short duration of recordings usually performed by athletes. As well, the sensitivity of rMSSD and SD1 to breathing patterns is very low compared to spectral indices (136), which is useful for day-to-day monitoring under spontaneous breathing conditions in athletes (148). Importantly, the day-to-day variations of time domain indices during short

(5-10 min) recordings in the field are likely lower than those of spectral indices (table 9.1). While the absolute magnitude of a CV is not an issue per se (see below), large CVs may lower the signal-to-noise ratio, which could compromise the sensitivity of the measures.

To conclude, the priority should be to collect resting HR and time domain HRV indices such as rMSSD. Despite large discrepancies in the literature (probably related to methodological issues), these indices, when interpreted correctly, are two promising tools for monitoring changes in general fatigue, fitness, and performance both over short- (159) and longer-term training periods (142).

## Exercise Measures

Cardiovascular-related measures that can be measured practically during exercise to gather insight into training readiness and/or fitness or fatigue include heart rate during exercise and the variability of heart rate (HRV) during exercise.



**Figure 9.2** Overnight hypnogram and heart rate (HR) patterns during two nights subjectively rated as very good (i.e., 5/5 on a 5-point scale) by a 25-year-old team sport athlete. Physical activity (no intense exercise) preceding each night was similar (resting day, i.e., 2809 versus 2826 Kcal, as measured by triaxial accelerometers) (21, 40). Note that while both sleep efficiency (SE) and total slow wave sleep (SWS) time are similar over the two complete nights (which is consistent with the subjective rating of the night), the actual sleep stage distribution and fragmentation between the first 4 h of the two nights differ markedly (i.e., greater proportion of SWS during the first part of night A, 43% versus 24%, and more arousals (Ar, changes in sleep stages), 59 versus 35). The greater sleep fragmentation during night A is associated with more HR arousals, which directly increases the number of rapid (beat-by-beat) changes in HR (reflected by both the power density in the high frequencies (HF) or the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals, Ln rMSSD) and, more importantly, the power spectral density in the low frequencies (LF), which, in turn, increases dramatically the LF/HF ratio. Compared to night B, this results in a paradoxical

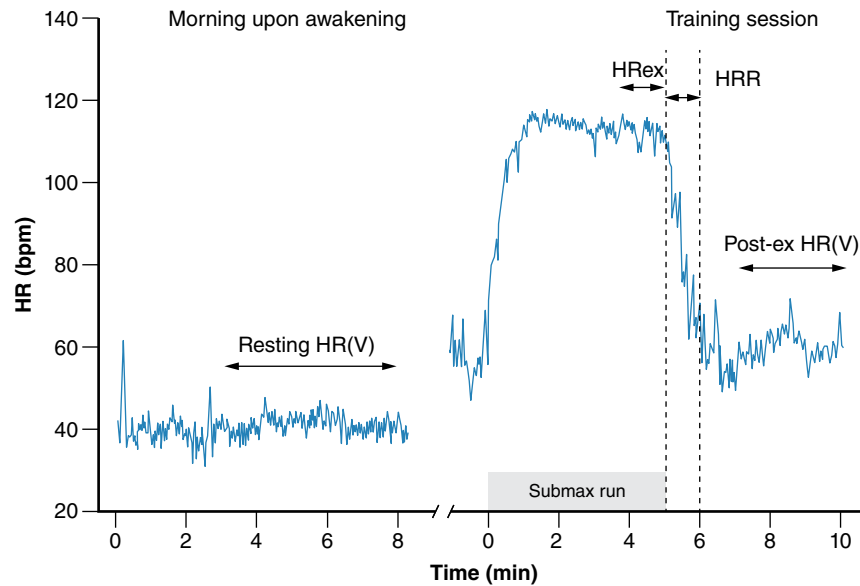
## Exercise HR

With resting heart rate, heart rate measured during exercise (HREx; figure 9.3) is probably the simplest measure to collect in individual athletes and does not require anything specialized such as beat-to-beat sampling. Depending on the intensity, exercising for about 3 to 4 min will generally allow heart rate to reach a steady state during submaximal exercise (56), and the last 30-60 s of data is generally used for analysis (e.g., (45, 46)). Since heart rate is closely related to  $O_2$  uptake during continuous exercise, HREx (when expressed as a percentage of maximum) provides a good marker of an athlete's exercise intensity (chapter 8), where lower relative heart rates at a given exercise intensity are generally indicative of a fitter athlete (114). Monitoring submaximal HR is common today in elite team sports clubs (35, 164). Many clubs use a 4 to 5 min steady state (~12-14 km/h) run-based monitoring on a weekly or monthly basis as an index of cardiovascular fitness. Generally speaking, an increase in heart rate should not be used as a marker of either fatigue and/or fitness impairment (43), although in the particular context of high-altitude camps, large increases (i.e., >4%) in HREx accompanying increased training loads could be predictive of illness (48). In summary, HREx is a highly recommended surveillance tool for monitoring positive aerobic-oriented training adaptations (43, 155).

Notwithstanding the ideal controlled scenario just mentioned, it unfortunately isn't what happens in

autonomic response, in which a greater vagal activity (Ln rMSSD) is superimposed to a greater sympathetic background (greater HR and LF/HF ratio). Autonomic coactivation may explain these observations (165), however, more likely it is a limitation of the HRV analysis methods to capture the actual ANS state during nonstationary periods such as sleep. Additionally, the greater Ln rMSSD and LF/HF values are inconsistent with the greater proportion of SWS during the first part of night A, since SWS is generally associated with both reduced Ln rMSSD and LF/HF (21, 132). Therefore, the calculation and interpretation of HRV indices over a large period of sleep, with no consideration of the actual sleep stage patterns, is particularly challenging and remains questionable for monitoring purposes. REM: rapid eye movements.

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**Figure 9.3** Example of the different HR recording conditions during the day. HR: heart rate; HRex: exercise HR; HRR: HR recovery over 60 s; HRV: HR variability.

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the life of most high-performance athletes and teams. Athletes and teams travel to different locations in the world at different times of the year, and environmental conditions change frequently. Because of the likely effect of heat on HR responses (especially during exercise performed outside), assessment of an athlete's fitness status must be made with consideration of the confounding influence that heat has on HR to avoid a false-positive assessment (result that indicates HR is elevated and fatigue is present). One way to avoid this is to perform the monitoring inside, where the environment can be controlled (47). However, in the likelihood of limited indoor space available for run-based monitoring, testing athletes on cycle ergometers is a good alternative (164), but perhaps less engaging for run-based team sport players. With most team sports, run-based monitoring on the pitch is the preferred option, so the temperature effect on HR must be known.

We showed that a 10°C drift in temperature leads roughly to a 1% shift in the fractional utilization of HRmax during a 4 min HRex monitoring run (figure 9.3) (102). From July and the summer training camp in hot conditions (temperature >35°C) to the cold winter of France (temperature <0°C), large variations in temperature can occur. Such variations independent of fitness changes can lead to changes of up to  $\pm 2\%$  in HR%, which are higher than the SWC and thus meaningful. To avoid misinterpretation

(i.e., players are assessed as unfit while the shift in HR% is due to hot temperature), it is necessary to adjust the HR values recorded based on outside temperature (figure 9.4b).

### HRV During Exercise

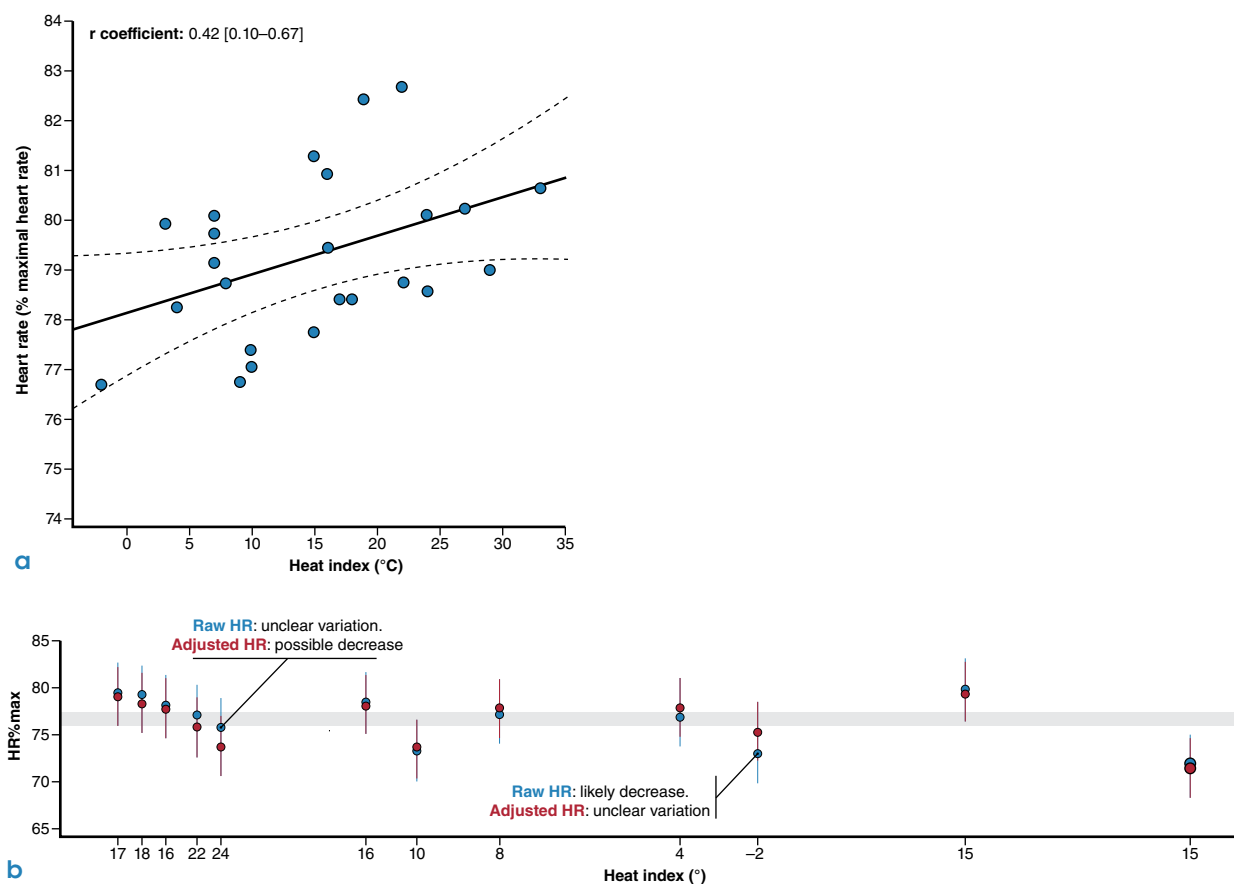
With respect to the measurement of HRV during exercise (figure 9.3), while attempts have been made to gain further insight into training-induced changes in fatigue and/or fitness (13, 101), there are still too many limitations with the method (24). Until new evidence is presented, the monitoring of HRV during exercise remains an interesting scientific toy but not a useful training monitoring tool.

### Post-Exercise Measures

With respect to gaining insight into training readiness and/or fitness or fatigue from cardiovascular-related measures that can be assessed practically after exercise, the main variables we should consider include the post-exercise heart rate recovery and the post-exercise heart rate variability (figure 9.3).

### Post-Exercise Heart Rate Recovery

Post-exercise heart rate recovery is reflective of a general hemodynamic adjustment in relation to body position, blood pressure regulation, and metaboreflex activity, which partly drive sympathetic withdrawal and parasympathetic reactivation (36, 59).



**Figure 9.4** (a) Relationship between heat index (index that combines air temperature and relative humidity to determine the human-perceived equivalent temperature, °C) and HR response during a 4 min submaximal monitoring run. Regression coefficients ( $r$ ) are presented as mean ( $\pm 90\%$  CL). (b) Intra-player changes in HR response (unadjusted (blue) and adjusted (red) based on heat index) to the 4 min submaximal monitoring run. During the fifth run, the unadjusted HR value suggests an unclear variation in fitness, however, the adjusted heat index ( $-24^{\circ}\text{C}$ )-related HR suggests a possible improvement (decreased HR). During the tenth run, the temperature was  $-2^{\circ}\text{C}$ ; unadjusted data suggest a likely increased fitness, while the variation appears in fact unclear when considering adjusted HR. Created with Tableau 10.2.

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We can measure HRR a number of ways, from the number of beats recovered within a given time (e.g., 60 s,  $\text{HRR}_{60\text{s}}$ ) to signal modeling via linear or (mono) exponential models (36). In practice, however, as good relationships between changes in simple HRR variables (i.e.,  $\text{HRR}_{60\text{s}}$ ) and changes in endurance performance and fatigue are reported (with the greater the increase in HRR, the better the performance) (59), HRR emerges as a relevant training surveillance tool for tracking positive changes in high-intensity exercise performance with individual athletes. However, increases in HRR have also been reported following functional overreaching (108),

which suggests that care should be taken when interpreting HRR changes and that other monitoring measures should be considered in parallel assessment (see limitations section). Nevertheless, changes in HRR have not always correlated with performance changes, in team sport athletes especially (43, 47), with correlations of lower magnitude than  $\text{HRR}_{\text{ex}}$ .

### Post-Exercise Heart Rate Variability

Post-exercise HRV (figure 9.3) has received some interest in sport science research, with the idea that it might offer deeper insight into training adaptations over and above that provided by resting HRV

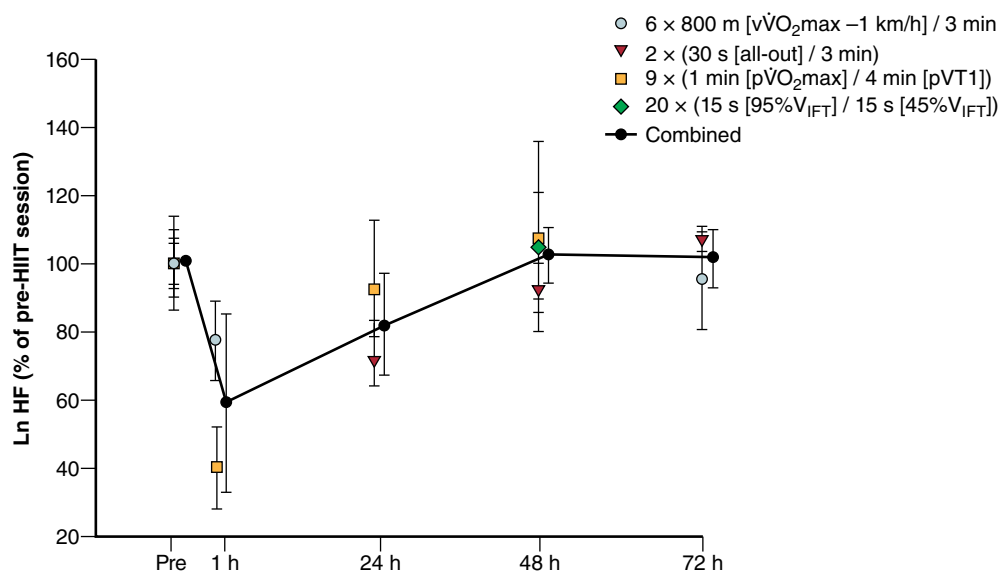
or HRR (171). The determinants of post-exercise HRV are multiple and include blood pressure regulation, baroreflex activity, and especially metaboreflex stimulation following exercise, which drives sympathetic withdrawal and parasympathetic reactivation (36, 159). High exercise intensities that drive blood acidosis and metaboreflex stimulation slow the HRR and lower the vagal-related HRV indices. Thus, to assess the true autonomic influences on HRV, independent of the metaboreflex stimulation, the use of submaximal exercise only ( $\leq VT_1$ ) has been suggested (36). Indeed, if exercise intensity is not low enough, post-exercise HRV is related to the exercise intensity only and additionally becomes less reliable (table 9.1). Thus, as post-exercise HRV offers no further insight over HRR, and is no more practical to implement in the field, being influenced by too many factors outside of even the exercise itself, its measurement is currently not considered of any use to monitor in athletes.

## Acute Responses

The natural day-to-day training load variations associated with the programs of most successful athletes are coupled by large daily variations of cardiac ANS activity (i.e., CV = 10%-20% for Ln rMSSD (38, 47)). In general, intense exercise acutely lowers vagal-

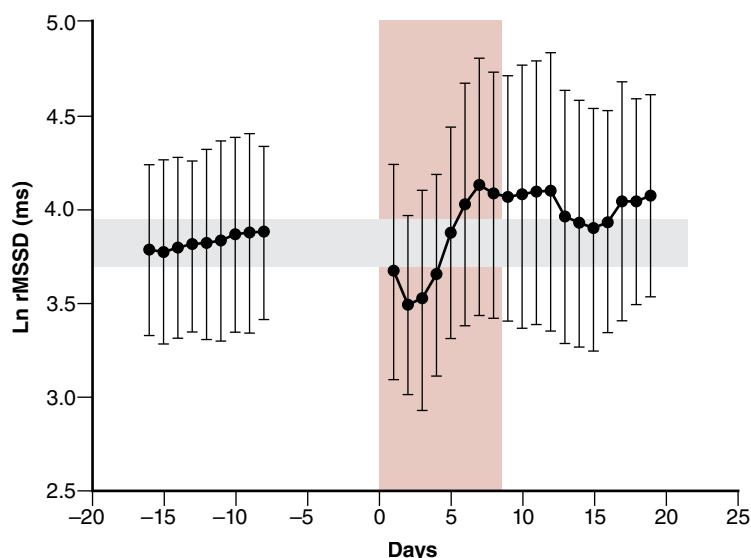
related HRV indices for 24 to 48 h (figure 9.5), likely a sign of the body's homeostasis restoration, and generally runs parallel to improvements in perceived levels of whole body recovery (159).

In line with such observations, HRV has been used experimentally to guide daily training content, in which subjects performed high-intensity training only when vagal-related HRV indices were at normal levels (98, 100, 159). While these HRV-guided training approaches have been shown to result in greater improvements in endurance performance compared to traditional programming (98, 100, 159), such an approach is not as simple as it sounds. For example, in association with heavy training loads in the heat, increased, not decreased, vagal-related HRV indices are observed in the following 24 h period, despite acute decreases in perceived wellness (47). Additionally, in the 2005 Marathon des Sables, an intense multiday desert race (running 253 km over 7 d in extremely hot conditions), after an expected initial drop in vagal-related indices during the first 3 d (22), we observed a clear increase in cardiac parasympathetic activity, which was opposite the increased fatigue and lowered running performance (figure 9.6). These inverse associations between vagal-related HRV indices and acute fatigue suggest that in addition to the load of the preceding session(s) (chapter 8), environmental conditions and hydration status



**Figure 9.5** Changes in the logarithm of the high-frequency component of heart rate variability (Ln HF) before (Pre) and following different HIIT sessions. pVT1: power associated with the first ventilatory threshold.

Adapted from James et al. (2002); Niewiadomski et al. (2007); Mourot et al. (2004); Buchheit et al. (2009).



**Figure 9.6** Average changes (90% CI) in the logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals (Ln rMSSD) measured at rest after awakening in six runners (4 M, 2 F,  $38.2 \pm 1.8$  yr,  $170.7 \pm 7.9$  cm,  $65.0 \pm 11.6$  kg,  $6.8 \pm 2.1$  h of training per week) before, during, and after the 20th Marathon des Sables (2005, 253 km in 7 d in a hot environment, with temperature sometimes exceeding  $50^{\circ}\text{C}$ ). The increase in Ln rMSSD after the fourth day of the race is unlikely reflective of general fatigue and fitness (which were reported to be deteriorated by the runners) but rather reflects changes in plasma volume due to the combination of exercise-induced hypervolemia (37) and heat acclimatization responses (47, 103).

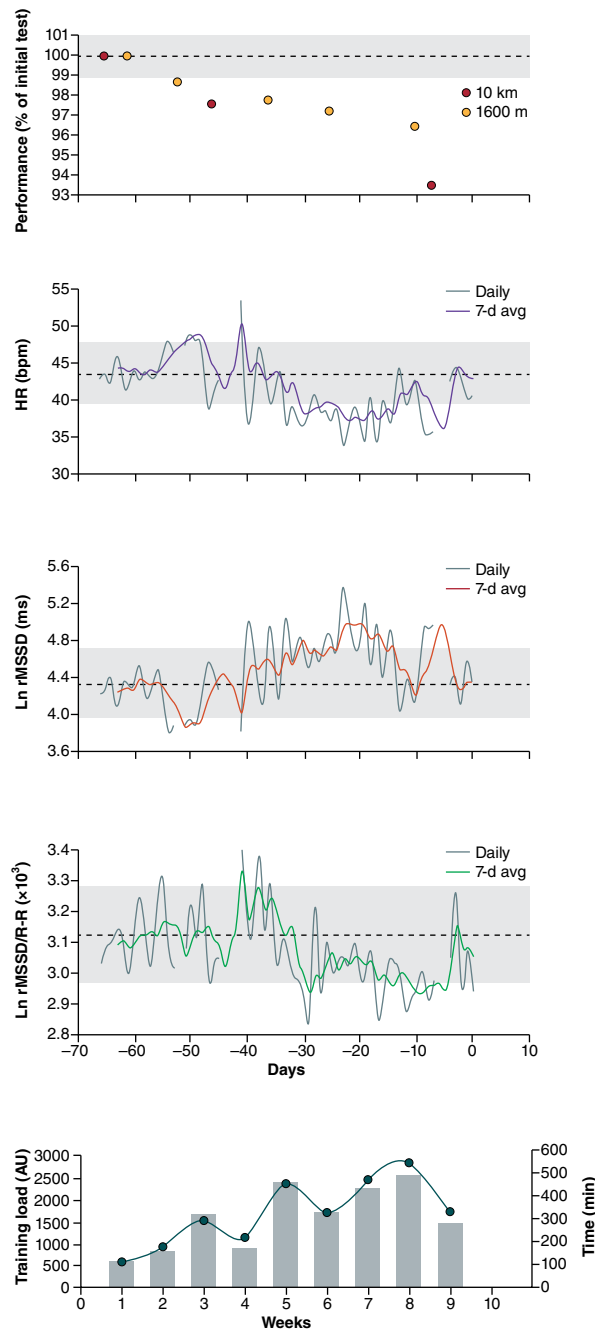
Unpublished data (Buchheit, Leprêtre, and Billat).

must also be considered for correct interpretation of the ANS response to exercise (47, 159). This is due to the fact that plasma volume increases occur in response to both intense aerobic-oriented exercise (71) and heat acclimatization (103), which tends to increase beat-to-beat HRV independently of clear changes in fatigue and/or fitness (37, 47, 158). Additionally, other factors that influence cardiac autonomic activity recovery time include hydrotherapy and sleep, which should also be considered when interpreting daily HRV changes (4, 160).

## Chronic Responses

HRV and HRR are known to fluctuate in accordance with training load periodization features, including volume, intensity, and overall load (46, 69, 93, 95, 117, 137, 138, 146). Moderate training loads are generally associated with increased vagal-related HRV indices, while high training loads are associated more with lowered vagal-related HRV indices (137, 138) and slower HRR (14). However, several studies in elite endurance athletes and/or athletes with a long training history do not always follow these trends,

with relationships in fact described more as bell shaped (93, 95, 110, 117, 142) (figure 9.7). In these particular endurance athlete subtypes, cardiac autonomic regulation is typically shown to improve during the first part of the training phase (e.g., building-up or extensive endurance phase, likely leading to functional overreaching), but decrease over the weeks preceding competition (i.e., tapering (91)). Since this particular pattern has been observed both in highly successful athletes (World and Olympic champions (93, 142)) and recreational athletes with long training histories successfully completing their first marathons (91, 95, 117), it may reflect an optimal training response. Importantly, performance has been reported to continually improve despite a lowered cardiac parasympathetic activity (figure 9.7), such that the relationship between these two variables appears reversed during the final stages of preparation. In support of this, while there are generally positive correlations between vagal-related HRV indices and markers of aerobic performance across large groups of athletes of varying training status (30, 46, 137), negative and large correlations are reported in homogeneous groups of well-trained



**Figure 9.7** Changes in running performance (best time over a self-paced 5×1600 m interval training session and an all-out 10 km run), resting heart rate (HR), logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals measured at rest after awakening (Ln rMSSD), the Ln rMSSD to mean R-R interval (Ln rMSSD/R-R), and training load (perceived exertion, CR-10 Borg scale×training duration (94)) and volume in a distance runner (32-yr-old,  $\dot{V}O_{2\max}=59$  mL·min<sup>-1</sup>·kg<sup>-1</sup>,  $v\dot{V}O_{2\max}=18$  km/h) over the 9 wk training program leading to his first

athletes (17, 42). Reasons to explain these training phase- or load-specific relationships are not fully understood, but might be related to the training intensity distribution during the different training phases (140, 142, 159), independent of fatigue. Indeed, low-intensity exercise, representing the greater proportion of training during high-volume training, is likely to acutely increase vagal-related HRV indices the day after exercise, while intense exercise, representing the greater proportion of training during tapering, lowers those indices for 24 to 48 h (159). This highlights the importance of considering training context (phases, training load and its distribution) when interpreting changes in HR-derived indices. Making inferences on the training status of an athlete based on HR measures without consideration of the training context is likely to be misleading. While assessing training load and training intensity distribution is straightforward in endurance sports (chapter 8), this is more difficult in team sports since the load arises from stress applied to countless biological systems simultaneously (see below). Therefore, whether the impact of training load (and intensity distribution) on

marathon (3 h, 15 min). The shaded areas represent trivial changes. Ln rMSSD and running performance change as a function of the training progression. While running performance improves continuously throughout, Ln rMSSD changes follow a bell-shaped relationship, with an initial increase followed by two consecutive reductions during the last phase of training. Interestingly, while the decrease in Ln rMSSD during week 7 to 8 is concomitant with a plateau of the Ln rMSSD/R-R ratio, the decrease observed during week 9 occurs with an increase of the Ln rMSSD/R-R ratio. This suggests that the decrease during week 7 to 8 is likely related to the saturation phenomenon (further increase in vagal activity), while that seen at week 9 is more likely related to decreased vagal activity and increased sympathetic activity (which might be required to reach greater exercise intensities during competition). This interpretation has important implications for using HRV to help guide the training process, which is only possible when using the combination of both indices.

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HR-derived indices described in endurance sport (140) is the same in team sports is not clear.

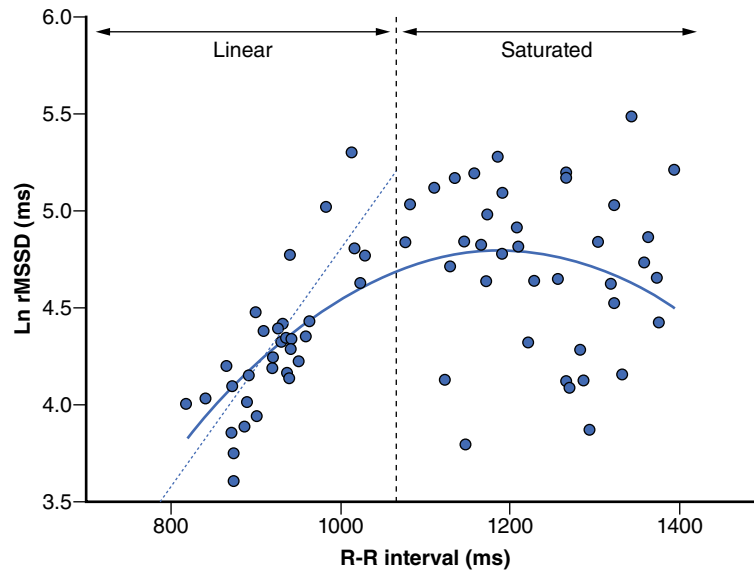
## Saturated HRV Profile

At first glance, athletes who show a low vagal-related HRV index are thought to be fatigued, under a heavy training load, or on the receiving end of some unexplained stress in their lives. But as we just alluded to, this is not actually the case in a number of athletes, even when such athletes are without fatigue, not training hard, and rested. Typically, the observation occurs in athletes with long training histories and is likely due to two different mechanisms. The first reason is a lowered global level of parasympathetic activity due typically to the polarized training load distribution (see chapters 3 and 8), reinforced by possible pre-competition stress (118, 124). This increased sympathetic activity is likely beneficial for HIIT performance, as it allows the attainment of greater levels of exercise intensity, often referred to as *maximal sympathetic mobilization* (79, 107, 133, 134). The second is what is known as the *saturation phenomenon* (99), which occurs independently of fatigue and/or sympathetic overactivity (40, 44, 99). This reduction in vagal-related HRV indices occurs with low HR levels and is actually the consequence of an increase, not a decrease, in vagal activity. The underlying mechanism can be understood if one understands any physiological saturation phenomenon. The one that is best understood today is that of insulin resistance, which happens when a person's cells become resistant to the hormone insulin. Ultimately the insulin has little effect on its main job of glucose disposal in cells at its setpoint concentration, thereby requiring more and more insulin to have any effect on lowering glucose concentration. Likewise with HRV, it is the saturation of the vagal-influenced acetylcholine in its receptors at the myocyte level, where this heightened level of vagal tone gives rise to a sustained parasympathetic control of the sinus node, thereby eliminating respiratory heart modulation, resulting in reduced vagal-related HRV indices (112). As the vagal-related HRV indices are more reflective of the magnitude of modulation in parasympathetic outflow as opposed to an overall parasympathetic tone per se (80), we observe a decrease in the HRV, despite increased vagal activity (i.e., typically shown as a high HRV response). Figure 9.8 shows an example in an athlete displaying the HRV saturated profile at low heart rates (higher R-R intervals).

The next point is critical for practitioners who want to interpret meaning from HRV measured in their athletes. Since the saturation phenomenon (figures 9.7 and 9.8) can confound the interpretation of training-induced adaptations, a number of approaches have been developed. These include using sitting (98), standing (46, 156), or post-exercise (45, 46) measures, which induce a minimal level of sustained sympathetic activity, constraining vagal activity below the tipping point of saturation (i.e., R-R interval >1000 ms (99, 139, 142)). When using the convenient resting supine or seated method, the only way of knowing whether a lowered vagal-related HRV index is related more to sympathetic overactivity versus saturation is to compare changes against resting heart rate. In practice with athletes, we recommend using the Ln rMSSD/R-R ratio (139, 142) (figure 9.8 and table 9.2). In the case of a sympathetic-driven lowering of Ln rMSSD, the R-R intervals should also be shortened (heart rate is higher), which means you see maintenance or an increase in the ratio. While moderate increases can sometimes be optimal (increased maximal sympathetic mobilization mentioned above), extreme increases in the ratio might reflect maladaptation to training, and in turn, lowered performance (142). In the case of saturation, the R-R is increased (heart rate is lower) and the ratio is substantially reduced (142). Whether saturation is beneficial for performance is unknown, as each athlete likely displays his or her own unique Ln rMSSD/R-R ratio profile (141), which may also be training cycle-dependent, suggesting that longitudinal monitoring over months or years is needed to maximize surveillance across each athlete (141). For an athlete showing a saturated profile during extensive training periods, a sudden loss of saturation suggests either increased readiness to perform (positive adaptation (142)) or fatigue/negative adaptation (15, 19)). To decipher between these two scenarios, practitioners need to consider the magnitude of the increase in the Ln rMSSD/R-R ratio (see above), alongside psychometric measures and/or face-to-face conversations with the athlete (47, 119).

## Interpreting Changes in Measures

In sports, so-called significant changes (i.e., changes determined based on null-hypothesis testing) in both performance tests and physiological variables, including HR-derived indices, GPS and accelerometry data, and wellness markers, are typically con-



**Figure 9.8** Example of the relationship between the R-R interval and the natural logarithm of the square root of the mean sum of the squared differences between R-R intervals (Ln rMSSD) in a subject with increasing bradycardia. Here, a saturation of heart rate variability is seen with long R-R intervals. Note how at shorter R-R intervals there is a linear relationship between Ln rMSSD (dotted line), which becomes disassociated as the duration of the R-R interval increases, indicating heart rate variability saturation.

Reprinted by permission of Springer Nature from D.J. Plews, P.B. Laursen, J. Stanley, A.E. Kilding, and M. Buchheit, "Training Adaptation and Heart Rate Variability in Elite Endurance Athletes: Opening the Door to Effective Monitoring," *Sports Medicine* 43 no. 9 (2013): 773-781.

sidered as having almost no practical relevance. Conversely, nonsignificant changes using null-hypothesis testing (P-value statistics) often have meaningful effects (87).

In order to avoid the embarrassing situation detailed in the sidebar *Appropriately Appreciating Signal with Respect to Noise: Avoiding Russian Roulette Programming*, we need a robust procedure. What actually matters to practitioners is whether the training-related change in a measure of relevance (such as HR) is actually important. Put another way, we want to know whether the magnitude of change

is essentially greater than the smallest practical or meaningful change or effect. Nowadays, we typically refer to this as the smallest worthwhile change (SWC) (10, 90). Additionally, when we compare the *average* changes in a performance or physiological variable of interest in a number of team sport athletes, across different metrics, and/or with heterogeneous between-athlete variances, the findings are often misleading (32).

For groups of athletes, an attractive option for assessing meaningful changes in any given variable we feel that matters and/or comparing changes in

## STATISTICS ARE OUR WEAPONS

"Statistics are our weapons" was first told to us by Dr. Nick Broad, 1974-2013. Dr. Broad was an English soccer nutritionist who worked for some of Britain's largest soccer clubs, including the Blackburn Rovers, Birmingham City, and Chelsea Football Club, and in Paris, for Paris Saint-Germain. Broad was a close friend of former Chelsea manager Carlo Ancelotti and graduated from Aberdeen University. He died on January 19, 2013, age 38, in an accidental traffic collision (169). We remember him also for his important message.

**Table 9.2** Guidelines for Interpreting Chronic Training Status Based on Different Scenarios in the Changes of Heart Rate–Based Measurements

| Resting HRV and HR |               |                           |                                                                                                                                                                                     |                                                                                   |                                                                                                                                                                                                                        |
|--------------------|---------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Changes in rMSSD   | Changes in HR | Changes in rMSSD/RR ratio | Occurrence                                                                                                                                                                          | Likely mechanisms                                                                 | Practical interpretation                                                                                                                                                                                               |
| ↑                  | ↓             | Maintained or moderate ↓  | Very frequent following a short-term training program in moderately trained athletes and/or during the building-up phase of elite athletes (high-volume and low-intensity training) | Increase in overall parasympathetic activity                                      | Coping well with training                                                                                                                                                                                              |
| ↑                  | ↑             | Large ↑                   | May occur at the beginning of a training block, likely observed in previously saturated athletes                                                                                    | Increase in sympathetic activity that reverses the saturation phenomenon          | 1. If occurring during tapering, increased readiness to perform<br>2. If not, accumulated fatigue                                                                                                                      |
| ↓                  | ↑             | Maintained or moderate ↑  | Frequent during tapering                                                                                                                                                            | Increase in sympathetic activity                                                  | 1. If occurring during tapering, increased readiness to perform<br>2. If not, accumulated fatigue                                                                                                                      |
| ↓                  | ↓             | Large ↓                   | Frequent in elite athletes or others with a long training history                                                                                                                   | Increase in overall parasympathetic activity that causes saturation               | 1. Elite athlete or athlete with long training history coping well with training, likely high-volume and low-intensity training<br>2. If prolonged and not reversed with tapering, can inform on an overtraining state |
| Exercise HR        |               |                           |                                                                                                                                                                                     |                                                                                   |                                                                                                                                                                                                                        |
| ↓                  |               |                           | Frequent, in relation to changes in training load                                                                                                                                   | Decrease in relative exercise intensity, plasma volume expansion                  | Cardiorespiratory fitness improvements                                                                                                                                                                                 |
| ↑                  |               |                           | Frequent, in relation to changes in training load                                                                                                                                   | Likely increase in relative exercise intensity or only reduction in plasma volume | Unclear; doesn't indicate decreased performance capacity                                                                                                                                                               |
| Post-exercise HRR  |               |                           |                                                                                                                                                                                     |                                                                                   |                                                                                                                                                                                                                        |
| ↑                  |               |                           | Very frequent following a short-term training program in moderately trained athletes and/or during the building-up phase of elite athletes (high-volume and low-intensity training) | Increase in overall parasympathetic activity                                      | 1. If occurring during tapering, increased readiness to perform<br>2. If prolonged and not reversed with tapering, can inform on an overtraining state                                                                 |
| ↓                  |               |                           | Frequent during tapering                                                                                                                                                            | Increase in sympathetic activity                                                  | 1. If occurring during tapering, increased readiness to perform<br>2. If not, accumulated fatigue                                                                                                                      |

Note that all changes need to be considered in relation to training context for appropriate interpretation. HR: heart rate; HRV: HR variability; rMSSD : square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals, reflecting cardiac parasympathetic activity; HRR: HR recovery.

## APPROPRIATELY APPRECIATING SIGNAL WITH RESPECT TO NOISE: AVOIDING RUSSIAN ROULETTE PROGRAMMING

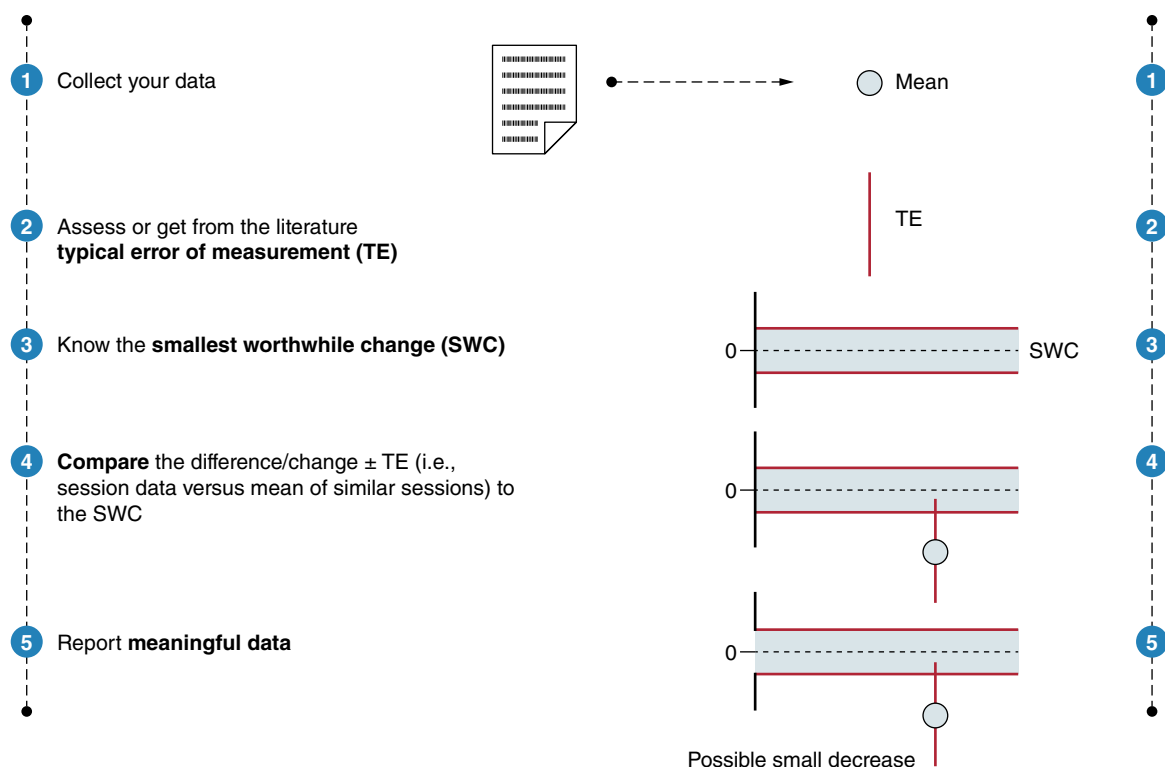
Martin Buchheit

One of the best examples of naïve data interpretation in the field that I can recall, which was unfortunately followed up with inappropriate decisions related to training, relates to a strength coach I used to work with long ago. The coach was fond of using technology to monitor workouts, though at the time we were in the infancy of understanding linear transducer technology and these types of innovative tools (see chapter 8). His main presumption (not a bad one actually) was that fatigued athletes needed to rest (or cycle easily on a stationary bike) and should avoid lifting weights at all. Conversely, fresh and fit athletes should double up their session efforts and go harder. While his idea had merit, his monitoring approach, being solely reliant on this technology, was excessive and, more importantly, lacked any sort of an established basis for making decisions. Sadly this cost him his job in the end. In fact, the transducer data were so unreliable at the time (notwithstanding the overall split-squat movement technique used to assess readiness) that athletes often had power data two times greater, or smaller, than those recorded in prior gym sessions. Immediate increases or reductions in power shown by the linear transducer machine, irrespective of magnitude, formed nonnegotiable indicators of training adaptation in his mind, with the resulting instructions to either do nothing or go hard. As the numbers were so detached from an athlete's perception of fatigue and readiness to train, his start-of-the-session split squat assessment became viewed a bit like Russian roulette or a random number generator. What my mate had forgotten in his monitoring approach was that what mattered most in any measured variable was the magnitude of change *relative* to the noise of the measurement. Without appreciating both means we'll often make the wrong decision.

variables of different metrics is using the concept of standardization (90). We can standardize any observed change by following Cohen's effect size principle, which is simply the change in the mean divided by the between-athlete SD of baseline test data. From this, the magnitude (standardized unit) can then be compared with magnitude-based thresholds that can be considered as small ( $>0.2$ - $0.6$ ), moderate ( $>0.6$ - $1.2$ ), large ( $>1.2$ - $2$ ), or very large ( $>2$ ) (90). Because the SWC is generally  $\sim 0.2$  standardized units, the assessment of substantial changes becomes relatively straightforward. However, because of individual responses in a team for example (evidenced by confidence intervals [CI] that will be large), meaningful inferences about the observed magnitude must be made to assess whether the effect is substantially beneficial or detrimental (clearly

greater than the SWC), trivial (clearly within the SWC), or eventually unclear (CI overlaps zero and/or the SWC). These inferences can be achieved using a specifically designed spreadsheet (i.e., post-only crossover for within-team changes (84)), where both the individual response (CI around the average change) and SWC are taken into account to make our final assessment (90).

In practice, however, practitioners need to monitor each athlete in isolation to individualize his or her training process, which requires a specific approach (88). As detailed in the following sections, the observed individual change in any given variable (such as heart rate) needs to be assessed in relation to its SWC (whether it is greater than the SWC, and if yes, by how much), while considering the possible error of measurement (see figure 9.9). The following



**Figure 9.9** Best-practice five-step statistical process for assessing changes in surveillance variables in an individual athlete.

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sections will detail how these two key elements can be determined.

### **Acknowledging Uncertainty Within the Measure**

When it comes to monitoring an individual athlete’s normal variation of measurement, or her within-athlete day-to-day variability, we use the typical error of measurement (TE), expressed as a coefficient of variation (CV) (85) (figure 9.9). The variability in each measure can be obtained either from the research literature or by using repeated measures of the variable collected in your athletes (88). The magnitude of these variations depends on both the variable itself (such as HR) and the recording conditions (i.e., HR measure CVs range from 3% to 190%, table 9.1) (149). For example, the CVs for time-domain HRV indices at rest are generally lower than those of spectral indices (especially for ratios, table 9.1) (5). Supine HR measures will show lower CVs than standing and exercise measures (149), while the CV for HRV during exercise is large, with values ranging from 120% to 190% (111, 120, 170). The

CVs for our recommended HRe<sub>x</sub> or HRR measures (3%-35%, table 9.1) are much lower than those of exercise HRV (5, 18, 62), and CVs decrease slightly with exercise intensity increases (104, 106). As only submaximal exercise can be used as a warm-up for most sports, using HIIT to derive less variable HRe<sub>x</sub> and HRR data is not practical. Moreover, while HR monitoring is often used as a diagnostic measure, it is unlikely that any coach would implement an intense exercise bout in an athlete already suspected to be overreached (chapter 7).

Importantly, it is not the absolute TE (or CV) of any measure that matters but the magnitude of its noise compared to the usual observed changes (signal) and the change that matters, or the one that has a practical effect (i.e., >SWC) (88) (figure 9.9). A measure showing a large TE but one that responds largely to training can actually be more sensitive and useful than one with a low TE but with a poor response to training. The greater the signal-to-noise ratio, the greater the sensitivity of the measurement. To understand the possible sensitivity of the different HR measures to positive adaptations to training,

training-induced changes in all HR variables (46) can be put in relation to the TE reported in the literature using a similar population (5). The comparison of the signal-to-noise ratio, calculated for each variable (range 0.7-1.6, table 9.1) suggests that HRex might be the most sensitive measure for tracking positive aerobic-oriented training adaptations. Importantly, however, these results often only apply within the context of the selected study (46) (e.g., short training period aimed at improving aerobic-related performance). The sensitivity of HRex to performance improvements can decrease over longer training periods (155), and whether it can be used to track other types of performance improvements (i.e., nonaerobically related) needs to be determined. Limited data exists on the sensitivity of HR measures for monitoring negative responses to training, and importantly, we know of no study in which all HR variables have been simultaneously reported, which would be required to compare the sensitivity of each HR index. In the study by Schmitt et al. (156), the signal (difference between fatigued and nonfatigued state)-to-noise ratio was shown as being greater for HRV indices (i.e., 3 for a vagal-related index) compared to resting HR (1.3) in the supine position. This difference in apparent sensitivity is consistent with the stronger trend shown toward nonfunctional overreaching for resting vagal-related HRV indices compared to resting HR ( $r=0.88$  versus  $0.81$ ) reported by Plews et al. (139). Considering that HRex does not always track impairments in physical performance (43), these data collectively suggest that HRV might be the more sensitive HR measure related to fatigue, but further studies are needed to confirm this.

The magnitude of the TE can also be reduced when repeating measurements, since noise will decrease by a factor of  $1/\sqrt{n}$  (i.e., 4 tests half the noise) (88). Interest in repeated measures has been shown in endurance athletes, where correlations with running performance in healthy (141) and overreached (110) athletes were observed using the average of at least 3 to 4 d of HR and HRV data, as opposed to isolated measures taken on single days (141, 144). However, we want a test with a TE that is lower or at least equal to the SWC; the ideal scenario is one with a TE less than half the SWC. Any change in the measure that is greater than the SWC will be almost certainly substantial (88). For other scenarios, practitioners are advised to use common sense and consider (visually) the degree of noise relative to the SWC band and whether it overlaps

(figure 9.9). When extreme precision in the interpretation is required, a specifically designed spreadsheet (assessing an individual (86)) can provide the practitioner with assessment of the chances (%) that the observed change is substantially greater or lower than the SWC.

### Determining the Smallest Worthwhile Change

While defining the SWC of any variable is a key aspect of the monitoring process (88) (figure 9.9), it is often overlooked in the literature (15, 59, 156, 166). A change that is clearly greater than the TE isn't necessarily practically meaningful. Indeed, it must be clearly greater than the SWC to arrive at this conclusion. Defining the appropriate magnitude of the SWC is complex, however, and will depend on training context, the adaptation type being monitored, and the variable itself.

For individual sports performance, expressing changes as a fraction of the within-athlete variation in performance is probably the best way of standardization, where we retain the ability to compare the obtained scores with magnitude thresholds. One-third of within-athlete variation in performance between competitions is generally considered as the SWC, while 0.9, 1.6, and 2.5 refer to moderate, large, and very large changes (88, 90). Simulations have shown that as much as  $0.3 \times CV$  performance improvements may result in a top athlete gaining one extra medal for every 10 races (89). However, when it comes to defining the SWC for individual changes in physiological markers, such as HR variables (that are not actually performance measures), the appropriate magnitude for the SWC is less straightforward (7). For HR(V) data, for example, in which day-to-day variability varies largely between individuals, using the within-athlete CV is likely the best option (25, 28, 29) (i.e.,  $SWC = \text{from } 0.5 \text{ (110) to } 1 \text{ (139, 142)} \times \text{individual CV}$ ). For performance variables in team sports (e.g., vertical jump, bench press), the magnitude of the between-athlete SD is related to group heterogeneity, which can be seen as a limitation; however, fractions (i.e., 0.2) of the between-athlete SD are still recommended (25, 28, 29, 46, 47). Since the magnitude of the between-athlete SD is related to group heterogeneity and it doesn't consider the variations inherent to repeated measures, using the (individual) CV (139, 142) as the reference for the SWC is intuitively a better choice. It is also worth noting that because of the training context dependency of HRV changes (143), the actual magnitude of the SWC may need to vary

throughout the training phases. HRV responses to either isolated sessions (159) or prolonged training phases (142) are also highly individual, so that the magnitude of the SWC for HR variables should also be individualized. However, there is no evidence that changes greater than any fraction/multiple of the CV are actually meaningful in practice with respect to actual changes in training status and/or performance. Therefore, to provide a starting point for understanding the magnitude of the SWC for each HR variable, group-average changes in HR variables following a successful endurance training period were extracted from the literature and related to the associated changes in performance variables for running (41, 43, 45-47) and cycling events (105, 114). The SWC for performance was considered as 1% for peak incremental test speed (43, 46) and 10 km running (46) and 40 km cycling (105, 114) time, and  $0.2 \times$  the between-athlete SD for the Yo-Yo intermittent recovery (41, 47) and repeated-sprint ability (45) tests. The HR-variable change corresponding to the performance-related SWC was then linearly extrapolated from the group-average  $\Delta$ performance/ $\Delta$ HR relationship in each study and considered as the SWC for the HR variable of interest (table 9.1). The SWCs obtained from each study were then averaged to produce a single SWC estimate for each HR measure. This group-average approach was preferred over within-individual modeling for two main reasons:

1. individual data points were not available for all studies, and
2. the noise inherent within HR measures can compromise the performance/HR relationships at the individual level.

Since the time course of HR can be bell-shaped when training is prolonged over several months, as with elite rowers (93, 141), triathletes (110, 139), or runners preparing for a marathon (95, 117) (figures 9.7 and 9.8), the use of the SWC provided in table 9.1 should be limited to moderately long training periods (build phases) and/or performance improvements within the range of those observed in Buchheit's study (46). While this bell-shaped time course likely reflects an optimal scenario (rowers were gold medalists at the Olympics and all runners ran their personal bests), the fact that HR variables returned to baseline levels by the end of the tapering phase and the lack of information on performance changes throughout the training programs means

that a clear estimation of the SWC in such cases isn't really possible. However, we can suggest that a 4% to 9% increase in vagal-related HRV indices during the build phase may be needed for athletes to compete optimally following the taper (93, 110, 139, 141) (figure 9.7). Because of the limited literature available on negative HRV responses to training and the lack of consensus on the expected relationships between HR variables and poor training status, the SWC magnitude for assessing negative adaptation remains. Until new evidence is provided, we suggest using, for convenience, the same threshold used for positive adaptations (4%-9%).

To close, we highlight what we feel is current best practice based on our previous work (25, 28, 29), and this framework is detailed in figure 9.9 for the practitioner.

**Making Decisions** Guidelines for interpreting changes in the different HR measures are shown in table 9.2. As emphasized throughout the book, training context is always crucial for interpreting changes correctly. When we monitor an athlete, there is always risk of finding either a false negative (no decision made when you should have) or a false positive (decision taken when you shouldn't have) error. In elite sport, such errors can be considered drastic in the context of performance optimization. Therefore, we recommend taking a conservative approach in our attempt at reducing the incidence of false negatives (48). It is better that we overcall and reduce training load (chapter 8) in athletes suspected of becoming tired or sick (chapter 7) even if they are not, versus the opposite. In this regard, using a lower SWC for our negative adaptation might be an option. Once a substantial impairment threshold in an HR index is identified, practitioners need to decide whether to make their decision immediately (i.e., alter training content in line with the diagnostic) or wait and see if the changes persist over 2 or 3 consecutive days (23). These rules of engagement can be either gradual or index dependent. For example, after observing a substantial decrease in resting rMSSD with the maintenance of training load, the training intensity distribution across a number of sessions can be shifted from intense to low, which should increase rMSSD without reducing overall load. If this marker was to further decrease after initial training adjustments, enforced rest could be applied (depending on associated fatigue; see neuromuscular and wellness sections below) and/or

specific reconditioning could be completed (in the case of detraining, in conjunction with other markers, see below). Finally, further analyses over larger numbers of athletes showing clear changes in training status are needed to determine the precision of sensitivity (proportion of true cases) and specificity (proportion of true noncases diagnosed as noncases) of each HR measure.

If a practitioner has the goal of using HR-derived measurements to help guide the daily training prescription, the only appropriate option really is the use of the morning resting HRV (98, 100, 159). Commercial smartphone technologies have made this very convenient for athletes, coaches, and scientists alike. The first advantage of the method is that there is often time for the data to be analyzed, typically before the training commences, so training can be modified accordingly, although this is not always the case for athletes who start training at the crack of dawn. Second, resting HRV measures can be repeated at any time (77) to assess the detailed recovery time course of the cardiac ANS following training (159). In contrast with exercise and post-exercise measures, which are generally performed during the session warm-up, there is no analysis time needed prior to training. Additionally, the data collection frequency to track ANS recovery time course can be limited to the times when the athletes exercise (e.g., no measure until later in the day (77)) (159).

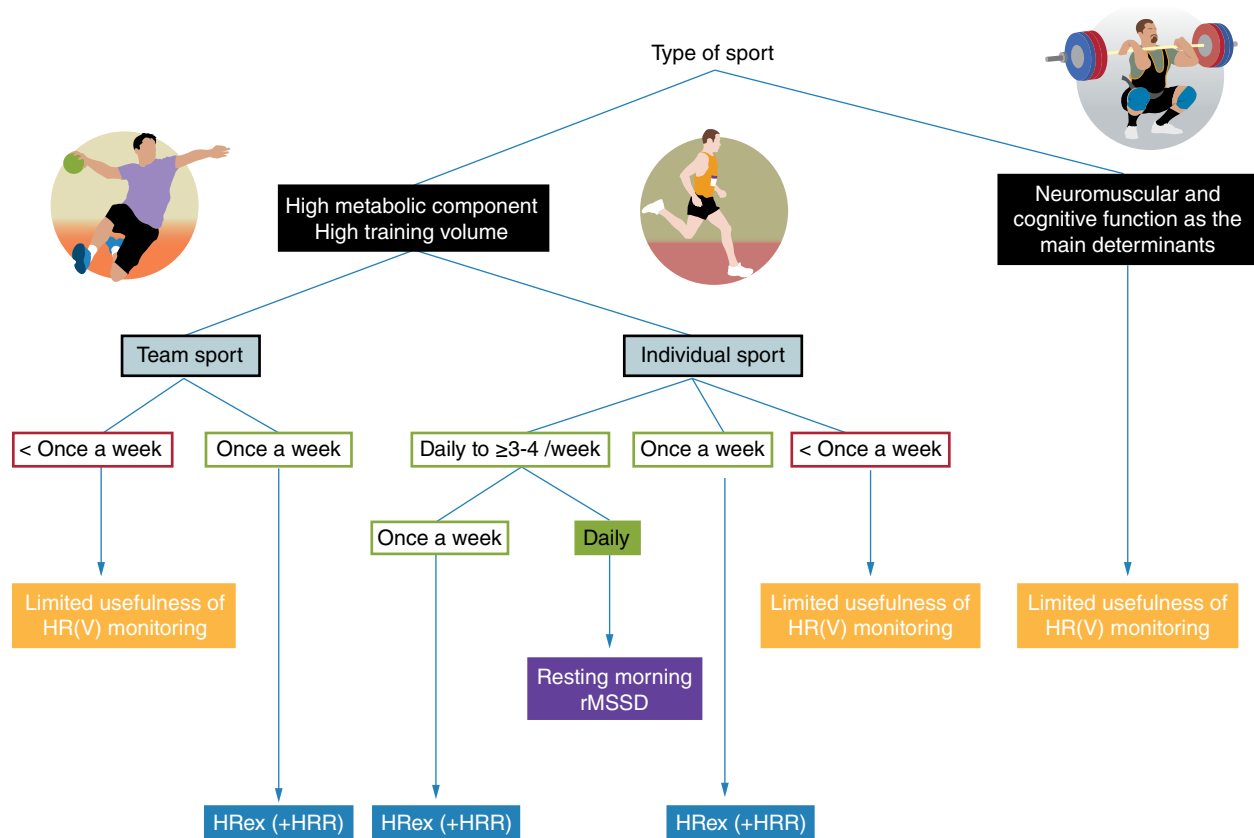
When it comes to selecting the most appropriate measure of moderate- to long-term training monitoring in athletes throughout the season, practitioners need to consider the balance between the power of a given measure (e.g., its relevance in relation to changes in fitness, fatigue, or both (32)) and common sense (e.g., ability to implement daily versus weekly). This directly determines the overall usefulness of the monitoring system. Consideration of the needs and the time constraints of each athlete, a measure that can be collected with high resolution (i.e., more frequently) is likely more useful than a possibly more powerful measure that can be collected only occasionally. Increasing measurement frequency allows us to define more accurately the normal ranges of individual variation and decreases the noise of measurement by a factor of  $1/\sqrt{n}$ , which in turn, improves our ability to adjust training content. The perfect scenario is to collect a combination of the most powerful measures daily.

Figure 9.10 shows a simple decision chart for helping practitioners select the most appropriate HR mea-

sure(s) to use in their monitoring system, based on their sport and measurement timing. The minimum number of daily recordings per week required to assess meaningful changes in training status from HRV is an important consideration with respect to an athlete's compliance. Endurance athletes might be advised to collect HRV on awakening 3 or 4 times a week (144) for the data to be averaged (110, 141, 142) and to supplement these measures once a week with HRex and HRR. Endurance athletes typically are at a greater risk of overtraining than team sport players, therefore requiring a more complete assessment. This population is also typically familiar with frequent monitoring and cope well with (almost) daily HR recordings. In contrast, implementing HR measures more frequently than once a week is unrealistic in the team sport setting, as home-based measurement in this population is outside of cultural acceptance and difficult for such athletes to implement. The only viable option in such groups is to collect HRex and eventually HRR each week on a standardized training day. Furthermore, the practitioner can implement the measurement of HRex into the team warm-up, which tends to be acceptable by coaches and fits well into the time available in elite team sports.

### Limitations of HR Measures

It is important to highlight here that HR measures cannot inform on all aspects of wellness, fatigue, and performance, especially as it relates to the response to HIIT, and especially the response to neuromuscular-related type 2, 4, 5, and 6 targets (31, 159). For example, while cardiac ANS recovery may align with muscular performance potential (57), there tends to be no association between this marker and blood creatine kinase or perceived muscle soreness following both a soccer game (41) and a 53 km mountain trail run (figure 9.7). Similarly, one has to wonder how cardiac ANS could possibly track metabolic recovery aspects such as glycogen resynthesis (31, 159). Finally, the time course of HRex and HRV adaptation during a training camp at 3600 m in highly trained young soccer players has been shown to be dissociated from both the changes in the RPE response to a submaximal run and psychometric measures of altitude tolerance (49). While mechanisms remain to be elucidated, the lack of sensitivity of HR measures to most neuromuscular, metabolic, or psychometric perturbations likely relate to the fact that HRV is only a marker of the *cardiac*



**Figure 9.10** Decision chart for the selection of heart rate (HR) measures based on sport participation and implementation possibilities. HRV: heart rate variability; Ln rMSSD: logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals measured after exercise; HReX: submaximal exercise heart rate; HRR: heart rate recovery.

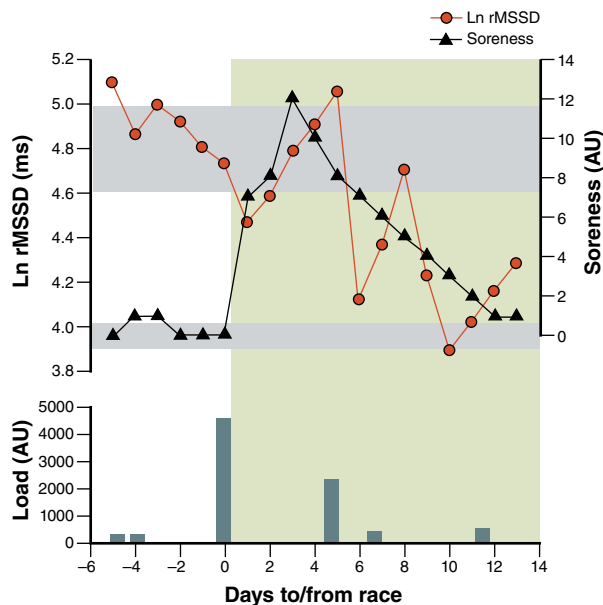
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ANS and nothing more (161) (see also figures 9.11 and 9.12). Therefore, the use of HR measures in combination with noninvasive neuromuscular-related markers found in GPS systems alongside countermovement jump scores (38, 58, 73) as well as general wellness and fatigue markers (83) may provide the complete solution for monitoring training status and response to HIIT load across most athletes and teams (47, 119). Therefore, assessment of changes in all relevant markers is needed to make the critical decision when it's required. The sport scientist needs to decide on the variable and physiological system that needs attention. As shown in figure 8.11, during congested periods of play in young soccer players, different time courses of adaptation may occur for individual players; fatigue can occur at the neuromuscular level without impaired cardiorespiratory fitness, and the converse. A last limitation of HR measurements is that athletes are generally required to use a chest belt, which is not

always convenient. There is however no doubt that future technological developments, where heart rate is collected via other sensor types (e.g., wrist-only monitors, clothes, sheets), will further increase the interest of HR monitoring in athletes.

## Summary of Tool 1

In summary, from a measure as simple as HR, a large number of indices related to internal metabolic, cardiopulmonary, and autonomic status can be gathered. In as little as 1 to 5 min, morning resting HR(V) can capture a snapshot of an athlete's cardiac parasympathetic activity on that day (142, 159), whereas a period of submaximal exercise HR (last min of a 4-5 min cycling/running) is additionally a useful monitoring variable. Resting HRV measures can be collected frequently to examine acute (daily recordings (159)) and chronic (at least 3-4 d per week (144))



**Figure 9.11** Time course of the logarithm of the square root of the mean of the sum of the squares of differences between adjacent normal R-R intervals measured supine at rest in the morning (Ln rMSSD) and perceived muscle soreness (Soreness, 0-10 scale) in a distance runner (32-yr-old,  $\dot{V}O_2\text{max}=59 \text{ mL}\cdot\text{min}^{-1}\cdot\text{kg}^{-1}$ ,  $v\dot{V}O_2\text{max}=18 \text{ km/h}$ ) before and after his first mountain trail (53 km, 4400 of positive ascent in 8 h, 22 min). Training/competitive load was provided as perceived exertion (CR-10 Borg scale)  $\times$  training/race duration (94). All sessions were run-based except the one at post-race +5, which was a bike session. The green area represents the post-race recovery period. Gray areas represent trivial changes for both Ln RMSSD and soreness. Note that post-race Ln rMSSD recovers within 2 d and rebounds above pre-race levels within 5 d. The recovery time course of Ln rMSSD unexpectedly mirrors the dramatic increase in post-race soreness. The impressive increase in soreness (12/10 scale!) is likely related to the fact that the runner was used to running on flat courses only (i.e., marathon training, figure 9.6) and not specially prepared for mountain running at that time.

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responses to training, and are likely sensitive to both positive and negative adaptations. Exercise HR may inform more on chronic and positive adaptations.

Correct interpretation requires each variable to be considered (1) in relation to the actual training

phase and/or load the athlete is under (Chapter 8), while (2) considering both the individual TE and the SWC. Finally, since HR measures cannot inform on all aspects of wellness, neuromuscular-related fatigue, and performance (31, 159), their use in combination with other markers, including GPS/accelerometers and psychometric/wellness markers (see Tool 2 and 3, respectively, below) may provide a complete solution to monitoring responses to training load in athletes (47, 119).

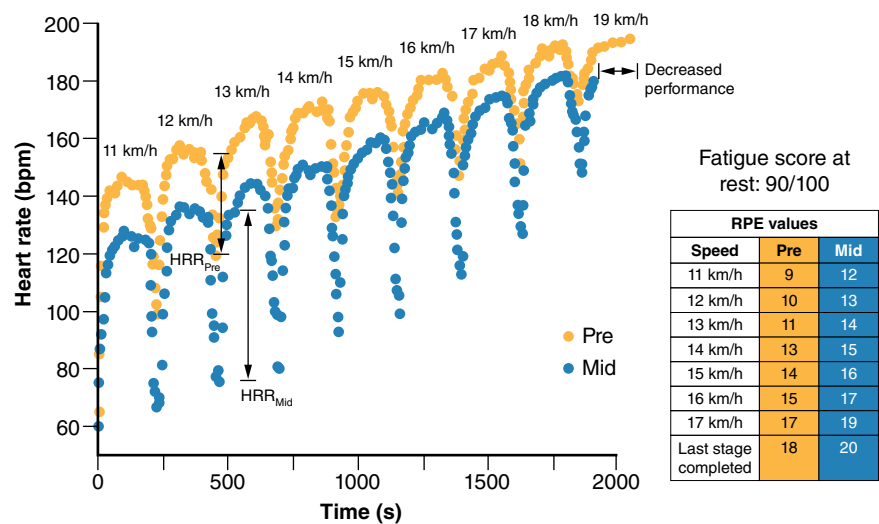
## Tool 2: GPS and Accelerometers

**With Mathieu Lacome and Ben Simpson**

Traditional methods used to quantify recovery and fatigue in athletes, such as maximal physical-performance assessments (figure 9.1), are typically not feasible for assessing fatigue status throughout competitive periods in team sport athletes, nor are they practical. Faster, simpler, and nonexhaustive tests, such as athlete self-reported measures (below), autonomic nervous system response via heart-rate-derived indices (above), and to a lesser extent, various protocols assessing neuromuscular function, may serve as promising tools to quantify and establish the big picture of an athlete's fatigue status in elite team sport athletes. While this section will focus on the needs within the team sport-specific context, there are nevertheless many concepts that are applicable to the individual athlete. Irrespective of the team versus individual athlete context, the implementation of appropriate, reliable, and sensitive measures of fatigue can provide important information to key stakeholders in any high-performance sport environment.

## Current Practices and Associated Challenges in the High-Performance Sport Setting

Since the early 2000s, there has been an exponential rise in research concerning training status monitoring in high-performance sports (6, 60, 61, 72), providing the foundation on which sport scientists may base their analyses. Extending from the aforementioned section on the key metabolic, cardiovascular, and autonomic monitoring aspects, the aim of the second section of this chapter is to attempt to look into the crystal ball of monitoring and outline the most relevant means of assessing neuromuscular performance aspects, including external



**Figure 9.12** Typical example of HR and RPE responses before and after the overload training period in a participant developing functional overreaching (i.e., decreased performance and high perceived fatigue). Note that the HR and HRR responses at the beginning of the test (i.e., low-intensity running) could suggest good adaptation to training when considered in isolation. However, the combination of these markers with RPE value analysis indicates the development of the functional overreaching state.

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force production and locomotor efficiency. While there is today evidence that neuromuscular-oriented performances, such as jump or groin squeeze tests or maximal sprints, can directly inform on the status of athletes’ neuromuscular function (157), there remain important limitations to their implementation in the elite setting, at least in some sports. For example, in elite soccer (chapter 30) or basketball (chapter 23), players are required to play every 2 to 4 d; it is simply not possible to perform such tests maximally between games. Therefore practitioners need to find other means of assessing the neuromuscular-related training status of athletes. Being personally involved in such a particular setting, we (102) have developed a specific monitoring approach based on various measurements that can be collected daily during normal training without the need for formal testing sessions and without requiring players to perform maximally. This section is a summary of our experience using GPS and accelerometry-based technology (102).

**GPS Tracking With Basic Metrics**

In high-level soccer clubs, it was shown that 40 of 41 surveyed teams collected heart rate and GPS data in every player during all field-based training sessions (2). Among the top-10 ranked variables used to quantify training load and response to load in

practice, distance covered in speed zones, accelerations, heart rate–related variables, and accelerometry metrics (e.g., PlayerLoad (8)) were the most frequently used markers. Referring to the classification of Gray in chapter 8, level 1 and 2 type data tends to be the most used in elite team sports. Since activity patterns of players are more influenced by contextual variables (e.g., rules, coach interventions, score lines, drills used) over player fitness status (53), locomotor-related variables (level 1 and 2) may not be the most vital markers used to directly monitor player training status (33). Additionally, given the diversity of team sport playing positions and/or player profiles that induce large between-player differences in locomotor activity, comparing locomotor performance between players is not very useful. It makes more sense to look at an individual player’s change in such variables. One potential option, although limited, is the use of standardized drills (such as D-1 game simulations, see chapter 30, Soccer, i.e., 9v9+GKs (33)) to assess changes in an individual player’s movement strategy (relative to himself) to gain insight into his training status. However, drill standardization is not always feasible in practice. For example, key players are often not available to train for various reasons, while congested fixtures (playing very often) means minimizing the team’s ability to perform a drill of interest.

To compensate for the limitations of level 1 or 2 variables in the absence of standardized drills, sport scientists generally examine a player's activity using two types of normalization:

1. Within-player analyses (102): A player compared with himself over multiple days, using historical data, i.e., intra-player trends.
2. Between-player normalization (102): A player compared with the team, with player locomotor activity systematically examined relative to the team or group of athletes.

These main two normalization aspects will be described in the following sections.

### Within-Player Analyses

Within-player analysis can include the daily readiness approach, as well as the acute–chronic workload ratio.

**Daily Readiness** With the daily readiness approach (table 9.3), practitioners can monitor changes and track trends in an individual player's activity using historical data to track eventual signs of acute fatigue or rebound in freshness and readiness to perform (33). In fact we can estimate a player's readiness to perform by comparing any activity on a given day across the range of an expected activity's intensity or volume (e.g., mean drill response  $\pm$  SD for a particular player for a D+3 and D-2 session; see soccer chapter). However, as differences in a

session's content (e.g., an added finishing drill compared to a usual D-1 session) or context (i.e., playing a Champions League game versus the bottom league team the next day in soccer) can create more of an effect on a player's locomotor activity versus changes in her fitness status per se. As a result, definitive conclusions become difficult to call.

**Acute–Chronic Ratio** Recently, the acute–chronic (A/C) workload ratio has received growing interest for monitoring injury risk (113, 126). The model does not aim to compare a specific session's varying locomotor response but rather tracks respective changes in a predefined acute (5-7 d) and chronic (21-28 d) load, using either internal and/or external load measures. Put simply, the A/C ratio gives us insight into the simple concept of “Am I doing too much work recently, relative to what my body is trained for?”

While the concept is promising, several limitations prevent a number of team sport clubs from fully utilizing the model (26). First, team sport scientists need to achieve the following:

1. Collect enough data (at least one full year of training load and injury data) to build an in-house specific model for the team.
2. Find the most meaningful A/C ratio time windows (e.g., 7/28 versus 6/21) that fit their sport's context (52).
3. Calculate the optimal A/C ratios for each measurement that matters (i.e., sRPE, high-speed

**Table 9.3** Reliability and Smallest Worthwhile Change for the Different GPS and Accelerometer Measures

| Variable                  | Technology            | Monitoring variable(s)       | Typical error, expressed as coefficient of variation        | Smallest worthwhile change                                  |
|---------------------------|-----------------------|------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| Daily readiness           | GPS                   | Readiness to perform         | Total distance: ~6%<br>Dist >14.4 km/h: ~18%<br>MechW: ~14% | Total distance: ~6%<br>Dist >14.4 km/h: ~18%<br>MechW: ~14% |
| Between-player Comparison | GPS                   | Readiness to perform         |                                                             | Total distance: ~2%<br>Dist >14.4 km/h: ~5%<br>MechW: ~5%   |
| HRA                       | GPS and HR            | Energetic cost of locomotion | 3%                                                          | ~1%                                                         |
| vL/FL                     | GPS and accelerometer | Locomotor efficiency         | 7.2 $\pm$ 4.1%                                              | 2%                                                          |
| Leg stiffness (K)         | Accelerometer         | Locomotor efficiency         | 11 $\pm$ 3.5%                                               | 4%                                                          |

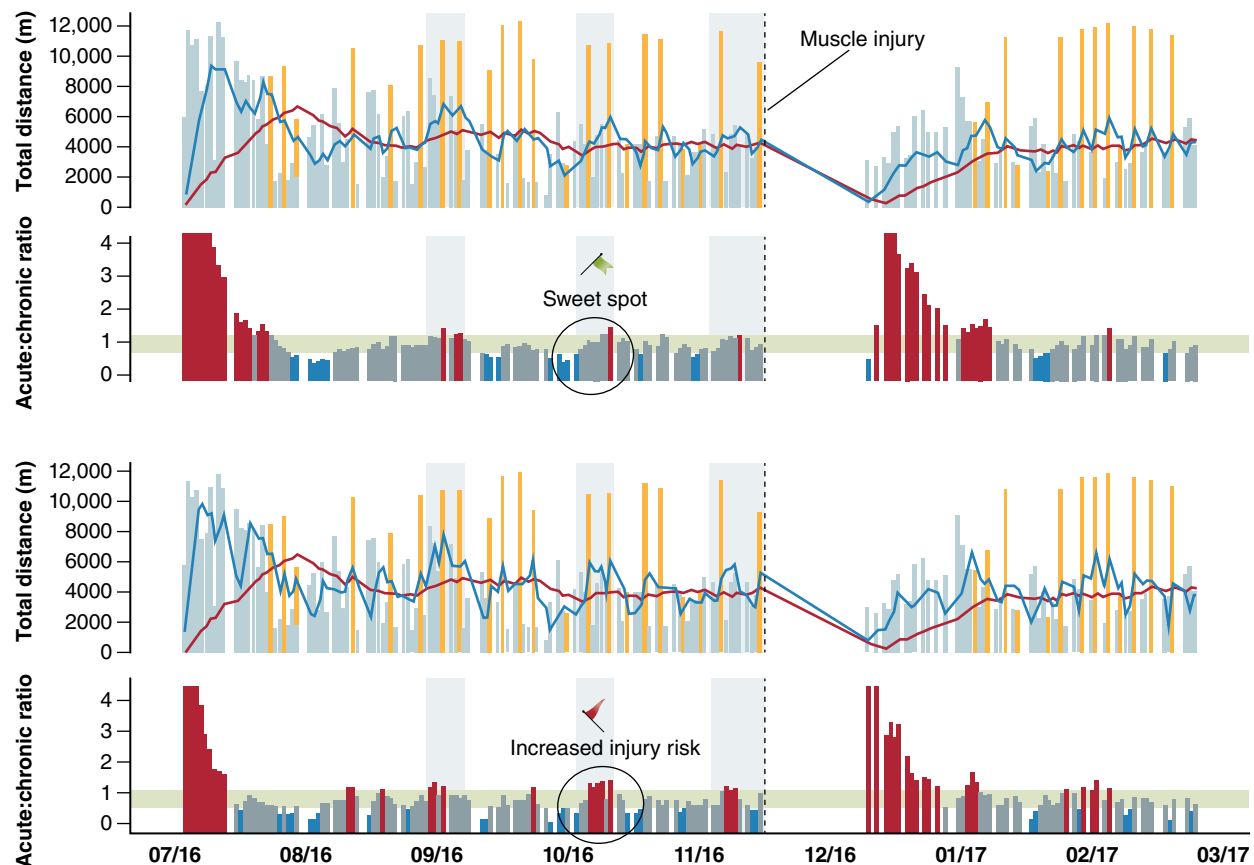
HRA is calculated as the difference between the HR responses to standardized drills such as small-sided games and the predicted HR for such drills based on individual multiples regression using GPS data as the independent variables. MechW: mechanical work; vL/FL: velocity load/force load.

running, mechanical work) since there are likely to be variable-specific sweet spots for decreased injury risk.

One important limitation that many elite team sports have to deal with are the periods when teams switch from club to international team and vice versa, which can sometimes be frequent. As one example, top soccer clubs in Europe have 60% to 75% of their players selected to perform on their national teams (A or U23-U19) at various times for international competitions. Because of little data-sharing between club and international team staff and the need for 21 to 28 d off to compute a chronic ratio, calculation of the A/C ratios is difficult (figure 9.13). With one international break every month from September 1 to mid-November, the use of A/C ratios will be

compromised during the first part of the season. While data can be estimated (26), this requires extensive work and adds uncertainty to the calculation. Figure 9.13 shows an example of an international soccer player. During the first part of his season, three international breaks required practitioners to estimate a substantial portion of the data (gray bars), lowering the confidence in the ratio calculated during the 3 mo period with the club. Additionally, the A/C window used (7:28 versus 4:18) means that we can have different interpretations of the same data (i.e., sweet spot versus increased injury risk).

Another limitation in our experience is the increasingly high number of false-positive outcomes that can be observed nearly every day (65), especially when using exponentially weighted moving averages (26) (e.g., ratio can be very high but players



**Figure 9.13** Measurement of total distance (m) in an elite soccer player over a 7 mo period. Acute (blue line) and chronic (red line) loads are calculated using 7 and 28 d periods (upper panel) and 4 and 18 d periods (lower panel), respectively. Dark gray zones represent international breaks when workloads were estimated based on the data obtained from the player's national team sports scientist. Total distance panels: gray bar, training session; yellow bar, match. Acute-chronic ratio panels: blue bars represent unloading (acute < chronic load) and red bars represent loading periods (acute > chronic load). Created in PowerBI (v2017-2.45).

remain injury free). As a result, players can directly discredit sports scientists who alarm coaches and medical staff on potential injury risk that doesn't manifest. Additionally, the fact that injuries can still occur in the 1 to 3 wk period (92) following a spike in load helps the defenders of the method to justify its interest to predict injury. Indeed, recurrent spikes in load are frequent in team sports, making it easy and convenient to find such occurrences to apply causation to the data set once a player is injured retrospectively (102).

Based on these highlighted limitations, we would encourage sport scientists willing to track (neuro-muscular) fatigue and injury risk to consider other methods of data analysis, such as the normalization of individual player locomotor activity.

### *Between-Player Normalization*

Player activity normalization (table 9.3) relative to the mean or median of the team or a group of specific players (same profile, same position) is another way to view data (equation 1). The main interest of this approach is that it is less likely affected by possible changes in session content for a given day (e.g., D+3, D-2 load), since all players can be considered. Additionally, the method can be used immediately on return from a national break, for example, since the metric doesn't rely on the previous week's load. Once historical data are available (e.g., previous season), insight into a player's fitness and/or early sign of fatigue can be gained while following the trends of these normalized data over several consecutive training days. In fact, sport scientists can use any variable (total distance, high-speed distance, accelerations, mechanical work-related variables/min) to construct similar models. Another possibility is the creation of a standardized value for each player based on the mean or median value of a typical drill used on a given day alongside the known standard deviation of the drill (equation 1). For example, following examination of the variability of most soccer drills we know, we have chosen to use only game simulations including goalkeepers (GS), possession-based games (PO), and some tactical and technical drills (102). Warm-ups have been removed due to the large between-player variability and overall low activity volumes. For our model, as the mean number of players is highly variable and generally less than 20 for the majority of the drills, the normalization is completed using the median value rather than the mean because it better represents the central tendency of small populations.

$$\text{Individual player readiness} = \frac{\text{drill locomotor value (TD)} - \text{drill group median (TDmed)}}{\text{standard deviation of the drill}}$$

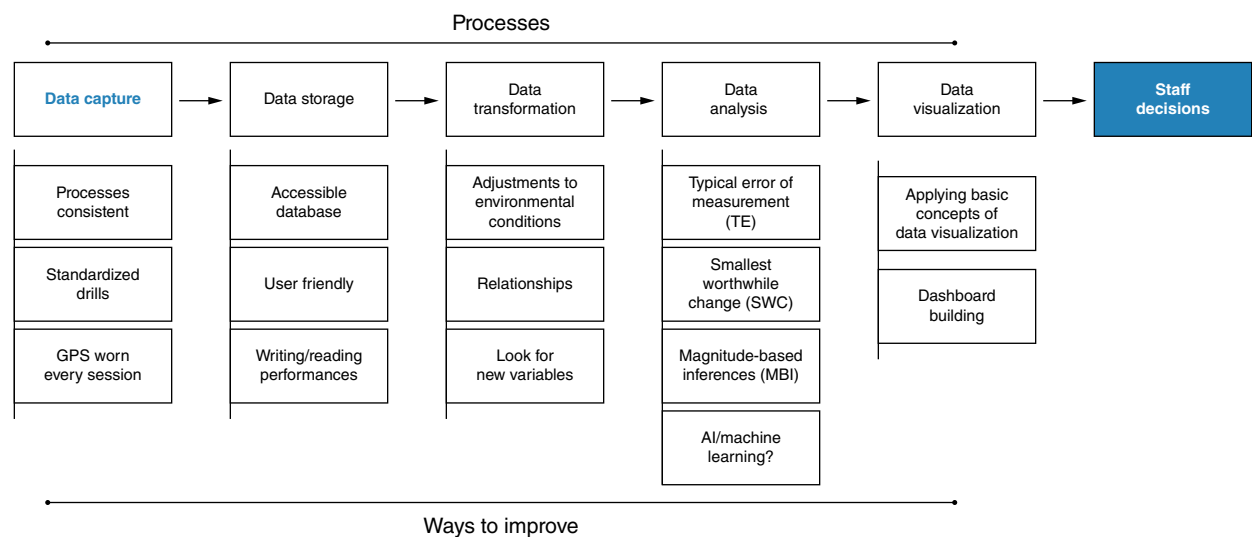
Despite its promise, there are still limitations to this methodology, especially when training a small group of players (top-up sessions, position-specific sessions), since the variability of the metrics likely increases with the lower number of players, or in player return to play (since the player trains by himself, no comparison can be made).

### **Optimizing the Model**

The quest of every practitioner working in elite sport is to continually search for an area of improvement in the collection, analysis, and reporting of data to the coaching staff (figure 9.13). To move forward, sport scientists can consider a simple yet effective framework for continual improvement by using:

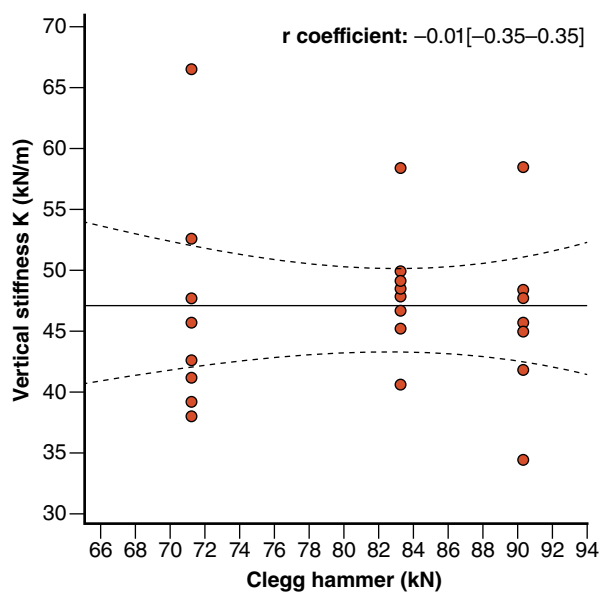
1. adjustment of previously used variables based on contextual factors to reduce noise/TE,
2. relationships between existing variables, and
3. integration of new and useful variables to strive for insight into a player's fitness, readiness to perform, and/or fatigue (figure 9.14).

For outdoor sports, weather and environmental conditions, as previously discussed (figure 9.4), contribute noise or variation to the measurements that ultimately become confounding factors in the model and need to be accounted for when interpreting data. Understanding the influence (or lack thereof) of the various contextual variables that can affect training load metrics is of great importance. With the recent use of advanced metrics (level 3 metrics (33)), sport scientists can record contact time, flight time, and/or leg stiffness (K) (34, 39), which means that understanding the player-pitch interaction is of interest. Yet little is known currently regarding how pitch surface (hardness, shear strength) can influence such metrics. While preliminary internal studies have shown that slight variation in pitch stiffness, as measured with Clegg Hammer (~70 to 85 kN), has no clear influence on metrics related to neuromuscular efficiency during running (K) (figure 9.15) (102), investigation into potential factors that could influence accelerometry data should be of interest in coming years. It is our responsibility as sport scientists to seek out every contextual variable that could increase the noise of the model, and in turn, lower our ability to detect meaningful changes in an athlete's



**Figure 9.14** Framework for continual optimization of training load response monitoring models.

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**Figure 9.15** Relationships between heat index (index that combines air temperature and relative humidity as an attempt to determine the human-perceived equivalent temperature, °C) and pitch hardness measured with Clegg Hammer (kN) and leg stiffness (K). Regression coefficients (r) are presented as mean (± 90% CL).

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fitness or fatigue state and/or a factor that would cause us to make the wrong interpretation.

### Building Relationships Between Existing Variables

In various team sports clubs around the world, compound metrics created by the combination of two or more variables are often used, requiring appropriate internal validation. For example, the ratio between velocity load (or m/min) and force load (or Player-Load) can provide a representation of the amount of force or ground impulses required per unit of displacement. Such a metric can be used to assess neuromuscular and running efficiency (the greater the ratio, the better the efficiency) and in turn neuromuscular freshness during standardized drills such as box-to-box runs or small-sided games (35, 39). While this metric requires proper validation, the concept in itself and preliminary results are promising.

To assess locomotor or work efficiency, and in turn infer potential fatigue and readiness to perform outside of the laboratory, sport scientists should assess the internal-to-external load ratios as a key marker of athlete readiness (figures 8.1, 8.2, and 9.1; table 9.3). In other words, simple internal-to-external load ratios can provide insight into an athlete’s training status and may be helpful in making decisions concerning training load adjustments. Using existing metrics to build relationships between internal- and external-related training load variables can be

a simple way to assess internal load relative to the external load as a cost/output relationship (50). For example, an increase in internal load relative to a standardized external load (e.g., monitoring run) can infer athlete fatigue or reduced fitness, while a decrease in internal load (e.g., a lower heart rate or perception of effort during a standardized external load) likely indicates that an athlete is gaining fitness and coping well with training. We used a ratio between RPE and relative total distance (RPE:m/min) to assess the overall acclimatization and fatigue trends during a training camp in a hot environment preceded by a long-haul flight (35). Akubat et al. (3) observed that the ratio between total distance (TD) and iTRIMP (chapter 8, compound measure based on HR) (TD:iTRIMP) during a standardized soccer-specific exercise was related to measures of fitness (vOBLA, vLT; chapter 3), suggesting that such a ratio might be used as a measure of readiness to perform.

As recently shown in Australian football (see chapter 21), measures of internal load (here RPE) are related to various external load metrics, while being highly individual (9). As such, individual relationships between internal load- and external load-related metrics could be assessed during a predetermined period to be used later as a prediction model. Subsequently, external load metrics (e.g., GPS metrics) can be used as independent variables to predict the dependent variable internal load (e.g., HR) after sessions. Therefore predicted HR responses could be compared to the real HR responses and a  $HR_{\text{real}}:HR_{\text{predicted}}$  ratio could in theory inform on whether the athlete was gaining fitness and coping well with training or alternatively gaining fatigue (see figure 9.16, panel Predicted vs Real (%)). Figure 9.16 summarizes these considerations and shows total distance covered in a player over 2 mo, across the start of his rehabilitation post-injury period to his full return to play. The different panels present (from top to bottom) total distance covered and the different ways to monitor injury risk or potential signs of fatigue: acute–chronic ratios, readiness index, and internal–external load relationships with  $HR_{\text{predicted}}$  versus  $HR_{\text{real}}$  comparisons.

## Potential for New Variables

With the ever-evolving advances in technology, a new array of GPS technology will soon be available (GPSport Evo, Statsport APEX, and Catapult G5, to cite a few), which incorporate improved GPS chips ( $\geq 15\text{Hz}$ ) and accelerometers ( $\geq 400\text{ Hz}$ ). Sport

scientists will encounter new challenges and opportunities in their athlete monitoring. Whereas Occam's razor (chapter 8) must still apply, level 3 (accelerometers, inertial measurement units)-type data will be easily available and likely more accurate than in the past. Thus, innovative and promising variables will be available for every session, and in turn, fatigue monitoring could become easier and more precise. With that in mind, practitioners should today consider the following opportunities related to force load and stride characteristic assessments in team sports.

### Force Load (fL)

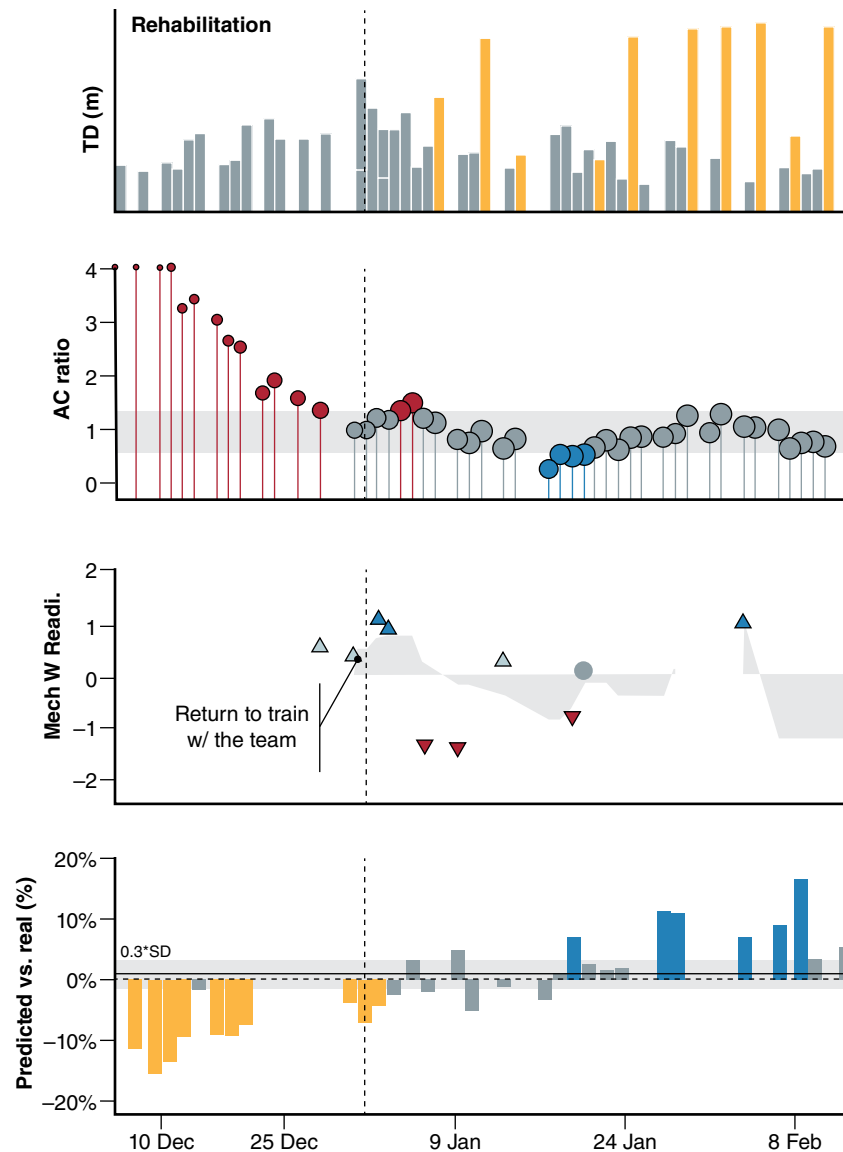
With the Athletic Data Innovation (ADI) analyzer, force load (table 9.3; (33)) refers to the sum of estimated ground reaction forces during all foot impacts, assessed via an accelerometry-derived magnitude vector. fL reflects only locomotor-related impacts and provides better estimates of overall footwork and impulses than total distance or PlayerLoad (8), especially when the session includes static movements and little displacements (i.e., rondos, free kicks).

During a given standardized drill, in relation to the average velocity (vL), vL/fL can be used to assess neuromuscular and running efficiency (the greater the ratio, the better the efficiency). Recently, the vL/fL ratio during box-to-box runs was shown to increase following soccer-specific endurance and speed sessions, suggesting a loss of efficiency in horizontal force application capability, likely due to the fatiguing effect of large amounts of high-speed running or training volume on posterior chain function (39). Also, moderate-to-large increases in vL/fL were observed 2 d after the end of an intense training camp in the heat, suggesting an increase in neuromuscular efficiency, which likely related to a rebound in the players' neuromuscular freshness (35).

fL can be compared between right and left legs during any locomotor actions (e.g., accelerating versus running at high speed, which is likely related to the use and potential weaknesses of different muscle groups (34).

### Stride Characteristics

Stride characteristics refer to contact and flight time calculated from accelerometry data. From these two variables, it is possible to calculate vertical stiffness (K, table 9.3), which has been shown to decrease substantially with neuromuscular fatigue (125). While the typical error for K is slightly greater when calculated in the field (i.e., box-to-box monitoring) compared to standardized runs on an indoor treadmill



**Figure 9.16** Training load (TD, m), acute-chronic ratio (AC ratio), readiness index (based on mechanical work, MechW Readiness), and HR response expressed as a percentage of predicted HR (Predicted vs Real (%)) in a typical player returning to training following injury. Vertical dashed line stands for the date of return to training with the whole group. TD (m): gray bar, training sessions; yellow bar, games. AC ratio: size of the circle related to chronic load (m); red circle, A:C > 1.5; blue circle, A:C < 0.8; gray area represents the theoretical sweet spot (0.8-1.5); A:C ratio > 1.5 during the rehabilitation phase due to preceding prolonged period without training. MechW Readiness: each form stands for one session standardized mechanical readiness (equation); blue form, MechW readiness > 0.2; red form, MechW readiness < 0.2; gray zone, rolling average over the last three sessions. Predicted vs Real (%): differences between GPS-based predicted HR and real session mean HR; orange bar, predicted < real, which means poorer than usual fitness; blue bar, predicted > real, which means better than usual fitness; in the absence of a clear value to define the smallest worthwhile change, the gray area was defined as  $0.2 \times$  between-player standard deviation. Following a prolonged period without training, A:C ratio progressively returns into a zone of reduced risk. At the same time, the difference between predicted and observed HR increases, which likely means that the player is gaining fitness. Created with Tableau 10.2.

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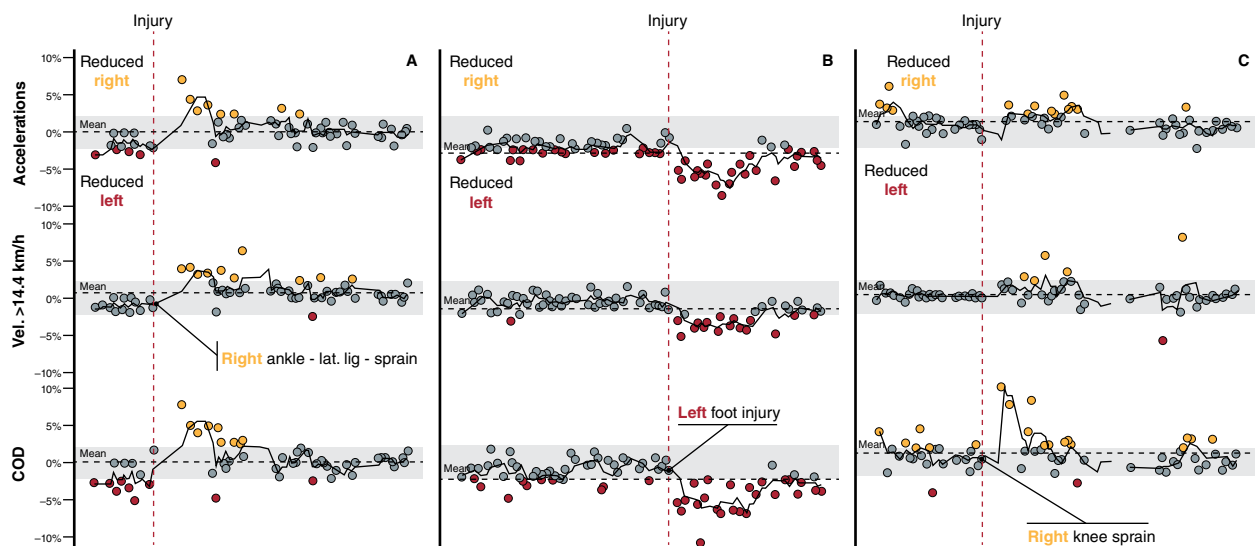
( $11\% \pm 4.5\%$  (34) versus  $6\% \pm 1.5\%$  (39)), the error remains small. The constant monitoring of stride characteristics during standardized running bouts in the field provides new perspectives for the monitoring of neuromuscular status under ecologically valid conditions.

Figure 9.17 shows how such variables can be used in practice. Panel A shows the symmetry calculated from the fL of all foot impacts when either running above 14.4 km/h, changing direction (COD), or accelerating ( $>2$  m/s) for all sessions of a player suffering from a tibiofibular inferior ligament sprain (right ankle). Following his injury, there was a clear force deficit on the left side, which progressively returned to baseline as long as the return-to-play program advanced. With these new metrics now available, especially due to the fact that the force load imbalances can be locomotor-phase specific (i.e., COD versus accelerations versus high-speed running), detailed injury-related patterns can be revealed. For example, in panel C, a player with a medial collateral ligament (MCL) sprain in the right knee presented with much greater strength imbalances during COD phases than during high-speed running and/or accelerations, likely due to the specificity of the strain associated with this injury. (The MCL is mainly

involved in protecting the knee against lateral force and is less affected by anteroposterior movements.) The diagnosis of strength imbalances is thus locomotion dependent, allowing sport scientists to complement and confirm the physician's or physiotherapist's manual testing protocols. This allows the provision of a fully functional diagnosis of the sprain or imbalance.

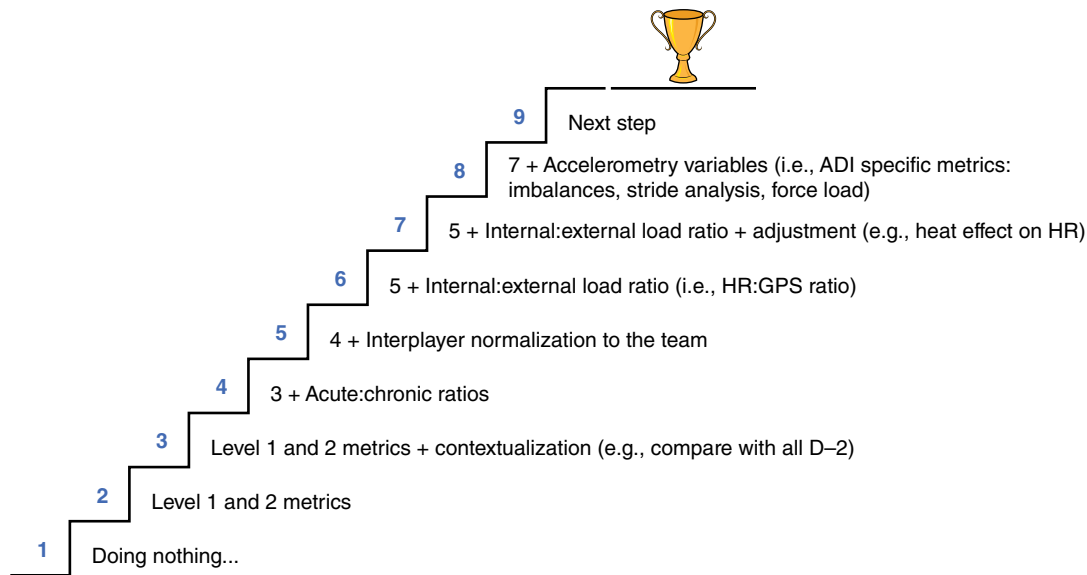
## Summary of Tool 2

When it comes to using technology to gain insight into an athlete's fitness, readiness to perform, and fatigue, we believe that a clear vision of the framework needed to develop, optimize, and improve the models over time is required. In fact, the measures themselves are unlikely to provide much information directly (in contrast to HR responses to a sub-maximal run, for example). It's more the way that data are analyzed and presented that make such measurements useful and relevant (27). In this regard, applying the right methods of analyses (stats, understanding the noise and SWC of each variable, table 9.3) and combining this information with those derived from other monitoring variables is what becomes useful (i.e., internal physiology with HR



**Figure 9.17** Force load symmetries in three players before their injuries and during the return-to-play period following tibiofibular inferior ligament sprain of the right ankle, left foot sprain, and medial collateral sprain of the right knee. The symmetry is calculated from the force load of all foot impacts during (from top to bottom) accelerations, running phase above 14.4 km/h, and changes of directions. Orange circle: right-leg force deficit  $>2\%$ ; red circle: left-leg force deficit  $>2\%$ ; red-dashed lines: injury date. Created in Tableau Software 10.2.

Reprinted by permission from M. Lacombe, B.M. Simpson, and M. Buchheit, "Monitoring Training Status With Player-Tracking Technology. Still on the Way to Rome," *Aspetar Sports Medicine Journal* 7, Targeted Topic: Football Science Evolution (June 2018): 54-66.



**Figure 9.18** The way to Rome: A summary of training load monitoring practices.

Reprinted by permission from M. Lacombe, B.M. Simpson, and M. Buchheit, “Monitoring Training Status With Player-Tracking Technology. Still on the Way to Rome,” *Aspetar Sports Medicine Journal* 7, Targeted Topic: Football Science Evolution (June 2018): 54-66.

measures, tool 1, and global health with wellness measures, tool 3).

A summary of the different training load monitoring options for practitioners, with specific reference to team sports, is shown in figure 9.18.

To this point in the chapter, we’ve assessed two of our three key families of importance for assessing performance potential or readiness to train, namely, our internal metabolic, cardiovascular, and autonomic marker of heart rate as well as our neuromuscular and external force efficiency markers through GPS and accelerometry systems. The final family of importance that we must acknowledge and respect in the context of HIIT and training load is the measurements that can offer insight into health and fatigue and what’s inside an athlete’s mind: the athlete’s psychology and wellness.

## Tool 3: Wellness

### With Anna Saw

*Wellness* is used in this book as an all-encompassing term to refer to an athlete’s physical, psychological, and/or social well-being. While some insight into physical wellness may be gathered from the biological markers already described, simply asking an athlete to report her perceived wellness may be the most holistic and meaningful approach (153). Wellness

measures typically take the form of a series of items with associated rating scales that can be either paper based or electronic, and which are completed on a regular, often daily, basis. The relative simplicity, accessibility, and universal nature of wellness measures make them a popular monitoring tool across all sports and participation levels. Even in well-resourced high-performance sport settings, wellness measures tend to be the most commonly adopted monitoring tool, used in approximately 80% of programs (162).

The use of wellness measures in practice was recently addressed in a systematic literature review, which provided strong empirical support for subjective wellness to reflect an athlete’s response to training (2, 153). Specifically, subjective wellness was shown to decrease in response to an acute increase in training load (chapter 8) and improve in response to an acute decrease in training load. The sensitivity and consistency by which wellness measures reflect the response to training load makes them ideal as surveillance tools.

Consider the following scenario in which a team of athletes attend a 2 wk training camp in the lead-up to a major competition. The first week is a solid training block, with more HIIT than the athletes had been completing in their home training environment. An acute increase in training load coupled with the additional stresses associated with the training camp environment is expected to elicit a decrease

in subjective wellness. Indeed, this pattern is seen across the team and the practitioner can be reassured that the desired training effect is being achieved. However, two individual athletes are flagged for deviating from this expected and typical team response. One athlete's wellness remains high, requiring the practitioner to investigate whether the athlete is not being sufficiently stressed by the training load. The other athlete has a disproportional decrease in wellness, requiring the practitioner to investigate whether the training load is excessively stressing the athlete or perhaps other nontraining factors are negatively affecting the athlete. The second week of the training camp is a taper week leading into competition. As such, subjective wellness is anticipated to improve. Similarly, the practitioner flags any athlete whose wellness does not improve as anticipated or deviates from the team's response.

This scenario illustrates how wellness measures can be used as an effective monitoring tool. However, achieving this textbook scenario in practice is no simple feat and requires both preliminary and ongoing efforts. Initially, careful consideration must be given to selecting a measure that has acceptable validity and reliability to enable the collection of meaningful data (figure 9.9). Data quality is then dictated by athlete compliance and honesty, which is influenced by how the measure is set up, supported, and used. Each of these factors is addressed in the following sections.

## Selecting a Wellness Measure

We have noted that wellness measures are both widely adopted in high-performance sport and well supported in the literature. However, we must acknowledge the divergence between research and practice when it comes to the particular measures employed. Experimental support for wellness measures is limited to established measures in the literature (153), yet practice tends to favor brief, custom measures (162). Recently, applied research using custom measures has noted similar responses to training as empirical measures (67, 163, 164), however, this is yet to be systematically evaluated. In order to avert the risk of less valid and reliable measures becoming repeatedly adopted and published (63), the following recommendations pertain to selecting an existing research-based measure.

Research-supported wellness measures assess one or more of the dimensions (overarching constructs) of mood, stress, recovery, symptoms, and emotions (table 9.4). Scales and items may relate to general

wellness (e.g., I was fed up with everything (96)) or be sport specific (e.g., I felt frustrated by my sport (96)). It may be argued that sport-specific items are more applicable to, and better received by, athletes. Alternatively, the argument that the training response also manifests as general wellness signs and symptoms suggests general items or a composite measure may be more appropriate for athlete monitoring.

Measures of subjective mood disturbance, perceived stress and recovery, and symptoms of stress have additionally been shown as being responsive to acute changes in training load (153). Therefore, there is no one dimension that may be recommended as the best option for athlete monitoring. Selecting multiple dimensions allows for variable athlete responses and avoids emphasizing one aspect of the training response over others.

The assessment of symptoms provides useful clinical insight into physical (e.g., headache, sore throat, digestive complaints) or behavioral (e.g., ability to work, eating habits) manifestations of the training response. However, in practice, it may be more advantageous to monitor mood disturbance and perceived stress and recovery in order to detect maladaptation prior to the appearance of symptoms (122).

The decision of which empirical wellness measure to implement will ultimately be dictated by the intended purpose and practicalities of the sport context (154). There should be a clear intention for how the data obtained will be used (i.e., to inform practices on a daily basis and/or with a longer-term perspective of improving understanding of athletic preparation for the future). Commitment of the coaching team to the process is of utmost importance, while efforts should also be invested in developing the buy-in of athletes, support staff, and the sports organization (152).

The selected measure must also be feasible in regards to the time and effort required of athletes and practitioners, in addition to the associated financial costs. It is important for sports to consider both the initial investment needed to set up the measure and monitoring process, ongoing costs of maintaining the measure (e.g., software updates), and interpreting and applying the data (e.g., practitioner time) as described below.

## Setting Up the Measure

Beyond the measure itself, the instruction given to athletes and the time and location of completion can affect the validity and reliability of responses. Athletes must understand the items and scales, however, it is

**Table 9.4** Empirically Supported Wellness Measures

| Measure                                                           | Primary dimension(s)   | General or sport-specific | Scales (n) | Items (n) | Time frame*    |
|-------------------------------------------------------------------|------------------------|---------------------------|------------|-----------|----------------|
| Profile of Mood States (POMS)(186)                                | Mood                   | G                         | 6          | 65        | 1 week         |
| POMS-Short Form (POMS-SF)(192)                                    | Mood                   | G                         | 6          | 37        | 1 week         |
| POMS-Abbreviated (POMS-A)(175)                                    | Mood                   | G                         | 7          | 40        | 1 week         |
| POMS-Adolescents/Brunel Mood Scale (BRUMS) (193, 194)             | Mood                   | G                         | 6          | 24        | 1 week         |
| Brief Assessment of Mood (BAM)(173)                               | Mood                   | G                         | 6          | 6         | 1 week         |
| POMS energy index(182)                                            | Mood                   | G                         | 2          | 15        | Now            |
| POMS training distress scale (190)                                | Mood                   | G                         | 2          | 7         | 1 week         |
| Multi-component Training Distress Scale (MTDS) (185)              | Mood, stress, symptoms | G,S                       | 6          | 22        | 1 day          |
| Daily Analyses of Life Demands for Athletes (DALDA)(191)          | Stress, symptoms       | G,S                       | 2          | 34        | Now            |
| Training Distress Scale (TDS)(174)                                | Symptoms               | G,S                       | 1          | 19        | 2 days         |
| Perceived Stress Scale (PSS)(172)                                 | Stress                 | G                         | 1          | 14/10     | 1 month        |
| Recovery-Stress Questionnaire for Athletes (RESTQ-Sport)(96, 179) | Stress, recovery       | G,S                       | 19/19/12   | 76/52/36  | 3 days/ nights |
| Acute Recovery and Stress Scale (ARSS)(176, 180)                  | Stress, recovery       | S                         | 8          | 32        | Now            |
| Short Recovery and Stress Scale (SRSS)(177, 180)                  | Stress, recovery       | S                         | 8          | 8         | Now            |
| Recovery-Cue (181)                                                | Stress, recovery       | G,S                       | 7          | 7         | 1 week         |
| Perceived Recovery Status Scale (PRS)(183)                        | Recovery               | S                         | 1          | 1         | Now            |
| Emotional Recovery Questionnaire (EmRecQ)(184)                    | Emotions               | G                         | 5          | 22        | Now            |
| Rating of Fatigue Scale (ROF)(187)                                | Fatigue                | G                         | 1          | 1         | Now            |

\*Time frame reflects response set of originally developed measure, however, this may be modified to meet requirements of context if permitted by the manual. G: general wellness; S: sport-specific wellness.

Adapted by permission from A.E. Saw, M. Kellmann, L.C. Main, and P.B. Gastin, "Athlete Self-Report Measures in Research and Practice: Considerations for the Discerning Reader and Fastidious Practitioner," *International Journal of Sports Physiology and Performance* 12 Suppl 2 (2017): S127-S135.

not necessary for this understanding to be universal among athletes. Athletes should complete the measure on their own and never discuss their responses with other athletes. As well, a standardized time and frequency for completion should be established (152), which will be related to the selected measure. However, some measures may need to be modified to suit the specific sport context.

A wellness measure can be set up as paper-based or electronic form or may be incorporated into one of the many commercial monitoring software platforms now available. Each mode has advantages and limitations, which should be weighed by the sport. Consideration should be given to the athlete completing the data, the practitioner overseeing compliance, and the individual- and group-level responses, with more in-depth insight completed as required. In

recent years, the accessibility of smartphones and the Internet has negated many of the previous limitations associated with electronic measures (152). Software applications now provide a simple and appealing solution for both athletes and practitioners.

It is important for software applications to be user friendly. Software that seems cumbersome, has glitches, or is not optimized for viewing on different devices will quickly deter use (152). As such, it is important that resources are invested in developing or purchasing a system that is simple, intuitive, and visually appealing. Ongoing investment is also necessary to ensure the software remains current and compatible with technological advances. Compatibility with other data sources, including training load, heart rate, and GPS measures (chapter 8 and above), is also desirable to facilitate integrated monitoring.

### **Creating a Positive Environment to Support Use**

The sporting environment will either serve to support or hinder the use of a wellness measure. In order to avoid the common pitfalls of poor compliance or dishonesty, it is important to invest in creating an environment in which athletes and practitioners buy in to the process, and athletes feel comfortable to disclose poorer states of wellness (152).

Prior to implementing the measure, practitioners should establish a clear purpose for the data, specify who will have access to the data, and how the data will and will not be used (including selection) (154). This process must be communicated to the athletes and reaffirmed through ongoing transparency and feedback. In sports programs in which several practitioners are involved in athlete monitoring, it is recommended that there is one key practitioner who coordinates input to ensure a consistent message is being fed back to the coaching team and athlete (129).

Introducing a wellness measure early in an athlete's career may serve to develop the habit of completion and also allow time for appreciation of the value of the measure to develop (12). Use of coercion or punishment to enforce compliance will undermine the long-term success of the measure and should be avoided (152).

### **Interpreting and Acting on the Data**

The interpretation of data requires the assessment of whether or not a change is meaningful. Interpretation of a meaningful change in wellness scores needs to take into account the individual's reporting habits. Some athletes habitually report within a very narrow range of values while others fluctuate considerably. Furthermore, the value an athlete considers as his normal may be the midpoint on the scale or at the lower end or upper end of the scale. To account for this, it is recommended that practitioners establish the CV for each individual from repeated measures between which no change in state is anticipated to have occurred (see previous section in this chapter). Once the individual CV is known, responses that exceed 1.5 to 2.0 times the CV may be flagged (85), however, this threshold has yet to be empirically established.

While attention typically is focused on detecting a meaningful change in an individual's data, it may be equally telling to detect a *lack* of change when one is expected. For instance, an athlete whose subjective

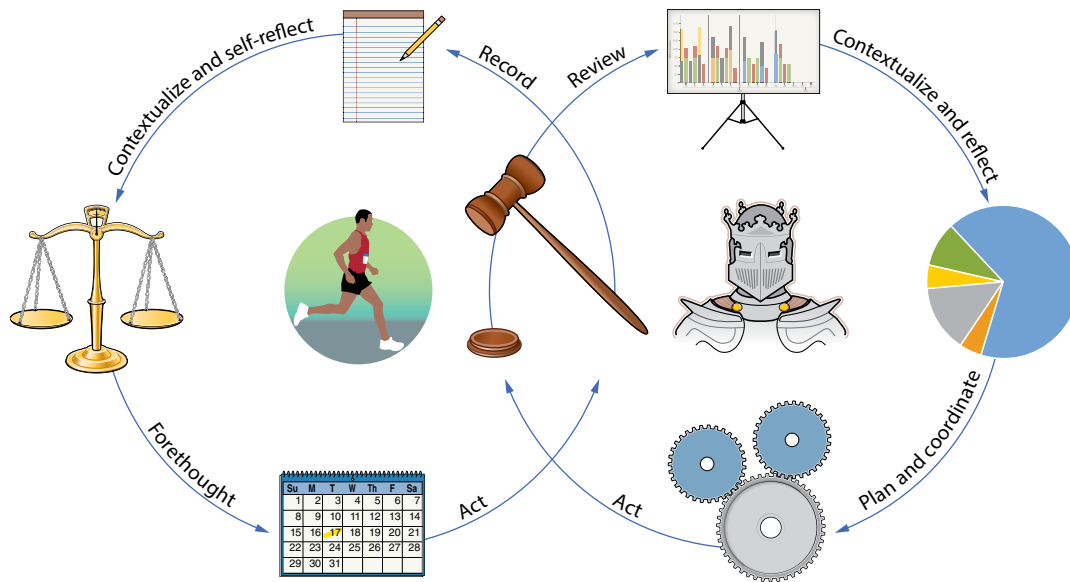
wellness remains stable rather than improving with a decrease in training load should be flagged, compared to their peers all being flagged for improving as expected.

Flagged wellness responses simply alert the practitioner to a potential issue and provide direction for further investigation (151). The practitioner must first consider the context of the wellness response (e.g., the training phase and expected wellness response, information from other monitoring tools, and knowledge of the athlete and her current situation) (151). If justified, follow-up actions include talking to the athlete and modifying training and recovery variables as appropriate. This process is illustrated in figure 9.19, which operates simultaneously with the self-monitoring (conscious or subconscious) of the athlete.

### **Additional Benefits Beyond Monitoring the Training Response**

When implemented effectively, wellness measures become a vehicle for improved communication between athletes and practitioners and also among practitioners. The measure provides an additional communication channel that enables athletes to disclose information they otherwise might not have. Athlete responses also serve as both a prompt and an invitation to discuss any concerns, whether it is the athlete approaching a practitioner or a practitioner approaching the athlete or other practitioners. Additionally, practitioners not working with the athlete daily can be kept informed on what the athlete is doing and how she is responding.

Strengthening relationships and collaboration among athletes and practitioners may serve to enhance the belief and confidence in a training program and that performance-related factors are accounted for and managed appropriately (72, 75, 151). Improved communication between practitioners facilitates the collaboration of different areas of expertise and perspectives to develop a more coordinated approach to athletic preparation. Furthermore, attention is directed to the importance of wellness for both performance and life. Indeed, use of a wellness measure may promote athletes to become more active and educated participants in their athletic preparation, both independently and by drawing on the support of practitioners (150). Furthermore, use of a wellness measure, independent of any practitioner involvement, is associated with improved athlete sporting self-confidence (150).



**Figure 9.19** Conceptual model of the process of incorporating a wellness measure into practice for both the athlete (left) and practitioner (right).

Adapted by permission from A.E. Saw, "Self-Report Measures in Athletic Preparation." PhD dissertation in School of Exercise and Nutrition Sciences. Deakin University, 2015.

### Summary of Tool 3

The relative simplicity and efficacy of wellness measures makes them a staple for every practitioner's surveillance toolkit. However, it is important for practitioners to appreciate that this efficacy depends on establishing strong foundations for data quality and continuing to invest in the process. As discussed in previous sections, incorporated wellness measures can aid in the interpretation of HR and GPS measures. Similarly, these objective measures can provide additional context to wellness measures to guide action for the coach and practitioner.

### Summary

In summary, the stress of training (chapter 7), quantified as training load (chapter 8), elicits a response to training that is highly individual. As this response to training is unique across athletes, understanding it is critical when assessing fitness, fatigue, and performance potential, determining adaptation, or assessing maladaptation or overreaching and injury risk. In this chapter, we have detailed the three distinct families of key training load response variables, each related to specific aspects of training status, which may all be useful in assessing performance potential or readiness to train. The categories, alongside the tools we believe best rep-

resent them, include the internal metabolic, cardiopulmonary, and autonomic status via heart rate; the neuromuscular or external force production and locomotor efficiency via GPS and accelerometry; and finally overall health and psychology from wellness questionnaires, describing the perceived global fatigue aspect of HIIT and the training load response.

Before finishing this chapter, there's one final point we'd like to highlight for tomorrow's sport scientist. In most elite sports (see chapters 11-30), the sport science department supports the coaching staff or performance manager, but the coach or coaches dictate the training program and, in turn, often the majority of the HIIT and training load (2). Effective communication to increase coach buy-in is now one of the more important, if not the most vital, of all soft skills needed for sport scientists working in the high-performance setting (27).

Today, coaches and performance managers need to delegate time to player demands, media requests, and sponsors, which highlights the importance of time-efficient practices for preparation, debriefing sessions, and/or the training plan. That said, it is unreasonable to think that most decision makers would spend more than 3 to 5 min reading any report; they simply don't have the time. Therefore, feedback must be accurate, straight to the point, and delivered in a timely manner.